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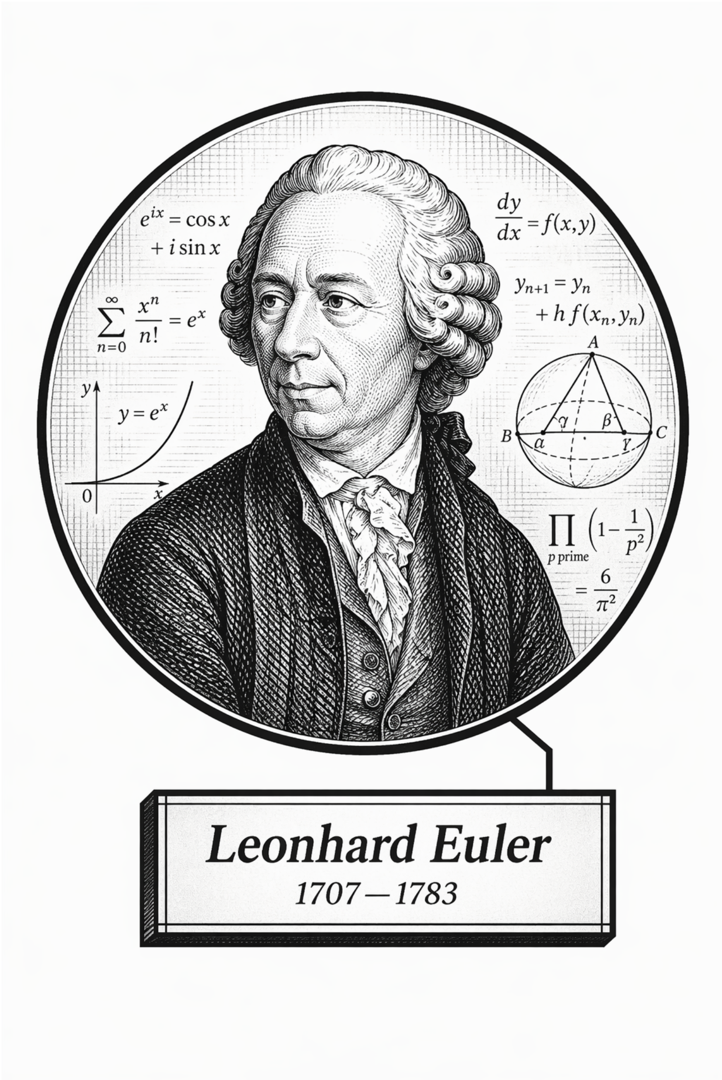

"I've got a number for you..."
— Leonhard Euler (probably)

Volume IV

Algebra and Abstract Structures

Self-Study Notes

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June 21, 2026

Chapter 1

Introduction to Algebraic Structures

algebraic-structures

Algebra $\rightarrow$ **Introduction to Algebraic Structures** $\rightarrow$ Algebra and Abstract Structures

[

Where You Are in the Journey]

Propositional Logic $\rightarrow$ Predicate Calculus $\rightarrow$ Sets & Functions $\rightarrow$ Proof Techniques $\rightarrow$ Real Analysis $\rightarrow$ **Intro to Algebraic Structures** $\rightarrow$ Linear Algebra $\rightarrow$ Topology $\rightarrow$ ...

This chapter begins with the idea of a mathematical structure and then builds toward familiar algebraic objects and number systems as examples of that idea.

1.1 What Is a Mathematical Structure?

Structures as Mathematical Objects

Remark (Exposition). An algebraic structure is not only a set. It is a carrier set together with specified operations, relations, distinguished elements, and laws. The signature records which symbols are available; a model interprets those symbols on a concrete carrier.

Carrier Sets

Definition 1.1.1 (Carrier Set). The **carrier set** of a structure is the underlying set on which the structure's operations, relations, and distinguished elements are interpreted.

Remark (Standard quantified statement).

$$\mathrm{CarrierSet}(A, \mathcal{A}) \iff A \text{ is the underlying set of } \mathcal{A}.$$

Remark (Definition predicate reading). A is the carrier set of $\mathcal{A}$ exactly when every symbol of $\mathcal{A}$ is interpreted over A.

Remark (Interpretation). In a group $(G, \star)$, the carrier set is G. In a ring $(R, +, \cdot)$, the carrier set is R. The additional symbols tell us what structure has been placed on that set.

Remark (Dependencies).

- TODO: formalize set as a reusable algebraic-structure prerequisite.

Operations

Remark (Exposition). Operations are function symbols interpreted as functions on the carrier set. Their arity records the number of inputs. Binary operations are central in algebra because they combine two elements of a carrier into another element of the same carrier.

Relations

Remark (Exposition). Relations are predicate symbols interpreted as subsets of finite Cartesian powers of the carrier set. Equality, order, divisibility, and equivalence are typical relations used to organize algebraic objects.

Distinguished Elements

Remark (Exposition). Distinguished elements are named elements of the carrier set. They often appear as constant symbols in a signature, such as 0 in an additive structure, 1 in a multiplicative structure, or e in a group.

Forward Reference: Model Theory

Remark (Exposition). Model theory separates syntax from interpretation. A signature names symbols; a structure interprets those symbols on a carrier set; axioms specify which structures count as models of a theory.

1.2 Relations

Binary Relations

Definition (Binary Relation)

Definition 1.2.1 (Binary Relation)**.** Let A be a set. A **binary relation on** A is a rule R that determines, for each ordered pair (a, b) with $a, b \in A$, whether a and b stand in the relation. We write $a\ R\ b$ when the relation holds, and $\neg(a\ R\ b)$ when it does not.

Remark (Standard quantified statement).

$$R \text{ is a binary relation on } A \iff R \subseteq A \times A.$$

Remark (Definition predicate reading). R is a binary relation on A exactly when every related pair has both coordinates in A.

Remark (Negated quantified statement).

$$\neg\,\mathrm{BinaryRelation}(R, A) \iff R \not\subseteq A \times A.$$

Remark (Interpretation). In set-theoretic language, a binary relation on A is a subset $R \subseteq A \times A$. The notation $a\ R\ b$ treats the relation as a predicate on pairs, which is the algebraic point of view used here.

Remark (Dependencies).

- Carrier Set

Properties of Relations

Definition 1.2.2 (Reflexive Relation)**.** Let R be a binary relation on A. The relation R is **reflexive** if every element of A is related to itself:

$$\forall a \in A, \quad a\ R\ a.$$

Remark (Standard quantified statement).

$$\mathrm{ReflexiveRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a \in A,\ a\ R\ a.$$

Remark (Definition predicate reading). R is reflexive on A exactly when every element of A has a loop to itself.

Remark (Negated quantified statement).

$$\neg\,\mathrm{ReflexiveRelation}(R, A) \iff \exists a \in A,\ \neg(a\ R\ a).$$

Remark (Interpretation). Reflexivity says that no element is excluded from relating to itself.

Remark (Dependencies).

- Binary Relation

Definition 1.2.3 (Irreflexive Relation)**.** Let R be a binary relation on A. The relation R is **irreflexive** if no element of A is related to itself:

$$\forall a \in A, \quad \neg(a\ R\ a).$$

Remark (Standard quantified statement).

$$\mathrm{IrreflexiveRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a \in A,\ \neg(a\ R\ a).$$

Remark (Definition predicate reading). R is irreflexive on A exactly when no element of A is related to itself.

Remark (Negated quantified statement).

$$\neg\,\mathrm{IrreflexiveRelation}(R, A) \iff \exists a \in A,\ a\ R\ a.$$

Remark (Interpretation). Irreflexivity forbids self-relation at every point of the carrier set.

Remark (Dependencies).

- Binary Relation

Definition 1.2.4 (Symmetric Relation)**.** Let R be a binary relation on A. The relation R is **symmetric** if

$$\forall a, b \in A, \quad a\; R\; b \Rightarrow b\; R\; a.$$

Remark (Standard quantified statement).

$$\mathrm{SymmetricRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a, b \in A,\ a\; R\; b \Rightarrow b\; R\; a.$$

Remark (Definition predicate reading)*.* R is symmetric on A exactly when every related ordered pair can be reversed.

Remark (Negated quantified statement).

$$\neg\,\mathrm{SymmetricRelation}(R, A) \iff \exists a, b \in A,\ a\; R\; b \wedge \neg(b\; R\; a).$$

Remark (Interpretation)*.* Symmetry says that the relation has no preferred direction.

Remark (Dependencies).

- Binary Relation

Definition 1.2.5 (Antisymmetric Relation)**.** Let R be a binary relation on A. The relation R is **antisymmetric** if

$$\forall a, b \in A, \quad a\; R\; b \wedge b\; R\; a \Rightarrow a = b.$$

Remark (Standard quantified statement).

$$\mathrm{AntisymmetricRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a, b \in A,\ a\; R\; b \wedge b\; R\; a \Rightarrow a = b.$$

Remark (Definition predicate reading)*.* R is antisymmetric on A exactly when two-way relatedness can occur only between equal elements.

Remark (Negated quantified statement).

$$\neg\,\mathrm{AntisymmetricRelation}(R, A) \iff \exists a, b \in A,\ a\; R\; b \wedge b\; R\; a \wedge a \neq b.$$

Remark (Interpretation)*.* Antisymmetry permits one-way comparison and self-comparison, but forbids nontrivial cycles of length two.

Remark (Dependencies).

- Binary Relation

Definition 1.2.6 (Asymmetric Relation)**.** Let R be a binary relation on A. The relation R is **asymmetric** if

$$\forall a, b \in A, \quad a\; R\; b \Rightarrow \neg(b\; R\; a).$$

Remark (Standard quantified statement).

$$\mathrm{AsymmetricRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a, b \in A,\ a\; R\; b \Rightarrow \neg(b\; R\; a).$$

Remark (Definition predicate reading). R is asymmetric on A exactly when every related ordered pair forbids its reverse.

Remark (Negated quantified statement).

$$\neg\,\mathrm{AsymmetricRelation}(R,A) \iff \exists a,b\in A,\ a\ R\ b\wedge b\ R\ a.$$

Remark (Interpretation). Asymmetry is stricter than antisymmetry and implies irreflexivity.

Remark (Dependencies).

- Binary Relation

Definition 1.2.7 (Transitive Relation)**.** Let R be a binary relation on A. The relation R is **transitive** if related pairs can be chained:

$$\forall a,b,c\in A,\quad a\ R\ b\wedge b\ R\ c\Rightarrow a\ R\ c.$$

Remark (Standard quantified statement).

$$\mathrm{TransitiveRelation}(R,A) \iff R\subseteq A\times A\wedge\forall a,b,c\in A,\ a\ R\ b\wedge b\ R\ c\Rightarrow a\ R\ c.$$

Remark (Definition predicate reading). R is transitive on A exactly when every two-step R-chain closes to a one-step relation.

Remark (Negated quantified statement).

$$\neg\,\mathrm{TransitiveRelation}(R,A) \iff \exists a,b,c\in A,\ a\ R\ b\wedge b\ R\ c\wedge\neg(a\ R\ c).$$

Remark (Interpretation). Transitivity is the chaining law behind order, equivalence, and divisibility.

Remark (Dependencies).

- Binary Relation

Definition 1.2.8 (Total Relation)**.** Let R be a binary relation on A. The relation R is **total** if every pair of elements is related in at least one direction:

$$\forall a,b\in A,\quad a\ R\ b\vee b\ R\ a.$$

Remark (Standard quantified statement).

$$\mathrm{TotalRelation}(R,A) \iff R\subseteq A\times A\wedge\forall a,b\in A,\ a\ R\ b\vee b\ R\ a.$$

Remark (Definition predicate reading). R is total on A exactly when every pair of elements is comparable by R.

Remark (Negated quantified statement).

$$\neg\,\mathrm{TotalRelation}(R,A) \iff \exists a,b\in A,\ \neg(a\ R\ b)\wedge\neg(b\ R\ a).$$

Remark (Interpretation). Totality rules out incomparable pairs.

Remark (Dependencies).

- Binary Relation

Equivalence Relations

Definition (Equivalence Relation)

Definition 1.2.9 (Equivalence Relation)**.** A binary relation R on A is an **equivalence relation** if it is reflexive, symmetric, and transitive. We write $a \sim b$ when R is an equivalence relation and the particular relation is understood from context.

Remark (Standard quantified statement).

$$\begin{aligned}\mathrm{EquivalenceRelation}(R, A) \iff\ & R \subseteq A \times A \\ & \land [(\forall a \in A)\, a\ R\ a] \\ & \land [(\forall a, b \in A)\, a\ R\ b \Rightarrow b\ R\ a] \\ & \land [(\forall a, b, c \in A)\, a\ R\ b \land b\ R\ c \Rightarrow a\ R\ c].\end{aligned}$$

Remark (Definition predicate reading). R is an equivalence relation on A exactly when it is reflexive, symmetric, and transitive.

Remark (Negated quantified statement).

$$\neg\,\mathrm{EquivalenceRelation}(R, A) \iff \neg\,\mathrm{ReflexiveRelation}(R, A) \lor \neg\,\mathrm{SymmetricRelation}(R, A) \lor \neg\,\mathrm{Transitive}$$

Remark (Interpretation). An equivalence relation on A is precisely the data needed to partition A into disjoint, exhaustive equivalence classes. Quotient groups, quotient rings, quotient vector spaces, and quotient topologies are all built by choosing equivalence relations compatible with the relevant operations.

Remark (Dependencies).

- Binary Relation
- Reflexive Relation
- Symmetric Relation
- Transitive Relation

Order Relations

Definition (Partial Order)

Definition 1.2.10 (Partial Order)**.** A binary relation R on A is a **partial order** if it is reflexive, antisymmetric, and transitive. A set A equipped with a partial order R is a **partially ordered set** or **poset**, written (A, R).

Remark (Standard quantified statement).

$$\mathrm{PartialOrder}(R, A) \iff \mathrm{ReflexiveRelation}(R, A) \land \mathrm{AntisymmetricRelation}(R, A) \land \mathrm{TransitiveRelation}(R$$

Remark (Definition predicate reading). R partially orders A exactly when it supports self-comparison, antisymmetric comparison, and chaining.

Remark (Negated quantified statement).

$$\neg\,\mathrm{PartialOrder}(R, A) \iff \neg\,\mathrm{ReflexiveRelation}(R, A) \lor \neg\,\mathrm{AntisymmetricRelation}(R, A) \lor \neg\,\mathrm{TransitiveRel}$$

Remark (Interpretation). Partial orders allow incomparable elements while retaining enough structure for comparison and chaining.

Remark (Dependencies).

- Binary Relation
- Reflexive Relation
- Antisymmetric Relation
- Transitive Relation

Definition 1.2.11 (Strict Partial Order)**.** A binary relation $<$ on A is a **strict partial order** if it is irreflexive, asymmetric, and transitive.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(<, A) \iff \mathrm{IrreflexiveRelation}(<, A) \wedge \mathrm{AsymmetricRelation}(<, A) \wedge \mathrm{TransitiveRelatio}$$

Remark (Definition predicate reading). $<$ is a strict partial order exactly when it compares without self-comparison, forbids reverse comparison, and chains transitively.

Remark (Negated quantified statement).

$$\neg\,\mathrm{StrictPartialOrder}(<, A) \iff \neg\,\mathrm{IrreflexiveRelation}(<, A) \vee \neg\,\mathrm{AsymmetricRelation}(<, A) \vee \neg\,\mathrm{Transitive}$$

Remark (Interpretation). Strict partial orders encode comparison without equality cases.

Remark (Dependencies).

- Irreflexive Relation
- Asymmetric Relation
- Transitive Relation

Definition (Total Order)

Definition 1.2.12 (Total Order)**.** A partial order $\leq$ on A is a **total order** if

$$\forall a, b \in A, \quad a \leq b \vee b \leq a.$$

A set equipped with a total order is a **totally ordered set** or **chain**.

Remark (Standard quantified statement).

$$\mathrm{TotalOrder}(\leq, A) \iff \mathrm{PartialOrder}(\leq, A) \wedge \forall a, b \in A,\ a \leq b \vee b \leq a.$$

Remark (Definition predicate reading). $\leq$ totally orders A exactly when it is a partial order and every pair of elements is comparable.

Remark (Negated quantified statement).

$$\neg\,\mathrm{TotalOrder}(\leq, A) \iff \neg\,\mathrm{PartialOrder}(\leq, A) \vee \exists a, b \in A,\ \neg(a \leq b) \wedge \neg(b \leq a).$$

Remark (Interpretation). Partial orders allow incomparable elements. Total orders require every pair to be comparable. A strict order can be recovered from a non-strict order by declaring $a < b$ exactly when $a \leq b$ and $a \neq b$.

Remark (Dependencies).

- Partial Order
- Total Relation

1.3 Operations

Binary Operations

Definition (Binary Operation)

Definition 1.3.1 (Binary Operation)**.** Let G be a set. A **binary operation** on G is a function

$$\star : G \times G \to G.$$

For $a, b \in G$, the element $\star(a, b)$ is denoted by $a \star b$.

Remark (Standard quantified statement).

$$\mathrm{BinaryOperation}(\star, G) \iff \forall a \in G\, \forall b \in G,\ a \star b \in G.$$

Remark (Definition predicate reading). $\star$ is a binary operation on G exactly when it combines any two elements of G and returns an element of G.

Remark (Negated quantified statement).

$$\neg\, \mathrm{BinaryOperation}(\star, G) \iff \exists a, b \in G,\ a \star b \notin G.$$

Remark (Interpretation). A binary operation takes two elements of the carrier set and returns another element of the same set. The codomain G encodes closure.

Remark (Dependencies).

- Carrier Set
- Function

Unary Operations

Definition (Unary Operation)

Definition 1.3.2 (Unary Operation)**.** Let S be a set. A **unary operation** on S is a function

$$f : S \to S.$$

Remark (Standard quantified statement).

$$\mathrm{UnaryOperation}(f, S) \iff f : S \to S.$$

Remark (Definition predicate reading). f is a unary operation on S exactly when it takes one element of S and returns one element of S.

Remark (Negated quantified statement).

$$\neg\,\mathrm{UnaryOperation}(f, S) \iff \exists x \in S,\ f(x) \notin S.$$

Remark (Interpretation). Unary operations appear throughout algebra: negation $a \mapsto -a$, multiplicative inverse $a \mapsto a^{-1}$, complex conjugation $z \mapsto \bar{z}$, and set complementation are all one-input operations.

Remark (Dependencies).

- Function
- Carrier Set

Nullary Operations

Definition (Nullary Operation)

Definition 1.3.3 (Nullary Operation)**.** Let S be a set. A **nullary operation** on S is the selection of a distinguished element of S.

Remark (Standard quantified statement).

$$\mathrm{NullaryOperation}(c, S) \iff c \in S.$$

Remark (Definition predicate reading). c is a nullary operation on S exactly when it selects a distinguished element of S.

Remark (Negated quantified statement).

$$\neg\,\mathrm{NullaryOperation}(c, S) \iff c \notin S.$$

Remark (Interpretation). In model-theoretic language, a constant symbol is interpreted as a nullary operation. The identity elements 0, 1, and e are typical algebraic examples.

Remark (Dependencies).

- Carrier Set

1.4 Laws of Operations

Associativity

Definition (Associativity)

Definition 1.4.1 (Associativity)**.** Let $\star$ be a binary operation on G. The operation $\star$ is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in G.$$

Remark (Standard quantified statement).

$$\text{Associative}(\star, G) \ :\equiv\ \forall a, b, c \in G,\ (a \star b) \star c = a \star (b \star c).$$

Remark (Definition predicate reading). $\star$ is associative on G exactly when any threefold product has the same value under either parenthesization.

Remark (Negated quantified statement).

$$\neg\,\text{Associative}(\star, G) \ :\equiv\ \exists a, b, c \in G,\ (a \star b) \star c \neq a \star (b \star c).$$

Remark (Interpretation). Associativity says that parenthesization does not affect the result of a chain of operations.

Remark (Dependencies).

- Binary Operation

Commutativity

Definition (Commutativity)

Definition 1.4.2 (Commutativity)**.** Let $\star$ be a binary operation on G. The operation $\star$ is **commutative** if

$$a \star b = b \star a \quad \text{for all } a, b \in G.$$

Remark (Standard quantified statement).

$$\text{Commutative}(\star, G) \ :\equiv\ \forall a, b \in G,\ a \star b = b \star a.$$

Remark (Definition predicate reading). $\star$ is commutative on G exactly when swapping the two inputs never changes the output.

Remark (Negated quantified statement).

$$\neg\,\text{Commutative}(\star, G) \ :\equiv\ \exists a, b \in G,\ a \star b \neq b \star a.$$

Remark (Interpretation). Commutativity says that the order of the two inputs does not affect the result. Associativity and commutativity are independent properties.

Remark (Dependencies).

- Binary Operation

Distributivity

Definition 1.4.3 (Left Distributivity)**.** Let $+$ and $\cdot$ be binary operations on A. Multiplication is **left distributive** over addition if

$$\forall a, b, c \in A, \quad a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark (Standard quantified statement).

$$\text{LeftDistributive}(\cdot, +, A) \iff \forall a, b, c \in A,\ a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark (Definition predicate reading)*.* Multiplication is left distributive over addition exactly when multiplying a sum on the left equals the sum of the two left products.

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{LeftDistributive}(\cdot,+,A) \iff \exists a,b,c \in A,\ a\cdot(b+c) \neq a\cdot b + a\cdot c.$$

Remark (Interpretation)*.* Left distributivity lets left multiplication pass through addition.

Remark (Dependencies)*.*

- Binary Operation

Definition 1.4.4 (Right Distributivity)**.** Let $+$ and $\cdot$ be binary operations on A. Multiplication is **right distributive** over addition if

$$\forall a,b,c \in A, \quad (a+b)\cdot c = a\cdot c + b\cdot c.$$

Remark (Standard quantified statement)*.*

$$\mathrm{RightDistributive}(\cdot,+,A) \iff \forall a,b,c \in A,\ (a+b)\cdot c = a\cdot c + b\cdot c.$$

Remark (Definition predicate reading)*.* Multiplication is right distributive over addition exactly when multiplying a sum on the right equals the sum of the two right products.

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{RightDistributive}(\cdot,+,A) \iff \exists a,b,c \in A,\ (a+b)\cdot c \neq a\cdot c + b\cdot c.$$

Remark (Interpretation)*.* Right distributivity lets right multiplication pass through addition.

Remark (Structural Significance)*.* Distributivity is the bridge between the additive and multiplicative operations in rings and fields.

Remark (Dependencies)*.*

- Binary Operation

Idempotence

Definition 1.4.5 (Idempotent Operation)**.** Let $\star$ be a binary operation on A. The operation $\star$ is **idempotent** if

$$\forall a \in A, \quad a\star a = a.$$

Remark (Standard quantified statement)*.*

$$\mathrm{IdempotentOperation}(\star,A) \iff \mathrm{BinaryOperation}(\star,A) \wedge \forall a \in A,\ a\star a = a.$$

Remark (Definition predicate reading)*.* $\star$ is idempotent exactly when repeating the same input does not change it.

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{IdempotentOperation}(\star,A) \iff \exists a \in A,\ a\star a \neq a.$$

Remark (Interpretation)*.* Idempotence says that self-combination stabilizes immediately.

Remark (Dependencies)*.*

- Binary Operation

Remark (Regeneration Note)*.* The prior working notes did not contain a clearly isolated idempotence discussion in the legacy Algebraic Structures chapter.

Cancellation Laws

Proposition 1.4.6 (Cancellation Laws in a Group)**.** *Let G be a group and let $a, b, c \in G$. If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$. Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{Group}(G, \star) \implies \forall a, b, c \in G,\ (a \star b = a \star c \Rightarrow b = c) \wedge (b \star a = c \star a \Rightarrow b = c).$$

Remark (Theorem predicate reading). In a group, a common left or right factor can be cancelled from an equation.

Remark (Negated quantified statement).

$$\mathrm{Group}(G, \star) \wedge \exists a, b, c \in G,\ (a \star b = a \star c \wedge b \neq c) \vee (b \star a = c \star a \wedge b \neq c) \implies \bot.$$

Remark (Interpretation). Cancellation is the algebraic effect of having inverses.

Remark (Dependencies).

- Group
- Inverse Element

Remark (Proof TODO). Regenerate this proof as a one-proof-per-file artifact.

1.5 Functions

Functions as Structure-Preserving Objects

Definition (Function)

Definition 1.5.1 (Function)**.** Let A and B be sets. A **function** from A to B is a relation $f \subseteq A \times B$ such that:

(i) $\forall a \in A,\ \exists b \in B,\ (a, b) \in f$;

(ii) if $(a, b_1) \in f$ and $(a, b_2) \in f$, then $b_1 = b_2$.

We write $f : A \to B$ and $f(a) = b$ in place of $(a, b) \in f$.

Remark (Standard quantified statement).

$$\mathrm{Function}(f, A, B) \quad \iff \quad \forall a \in A,\ \exists!\, b \in B \text{ such that } (a, b) \in f.$$

Remark (Definition predicate reading). f is a function from A to B exactly when every input in A has exactly one output in B.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Function}(f, A, B) \iff \exists a \in A \text{ with no output in } B \vee \exists a \in A \text{ with two distinct outputs in } B.$$

Remark (Interpretation). A function assigns exactly one output in B to every input in A. In algebra, functions become structure-preserving objects when they are required to respect specified operations, relations, or constants.

Remark (Dependencies).

- Binary Relation
- Domain and Codomain

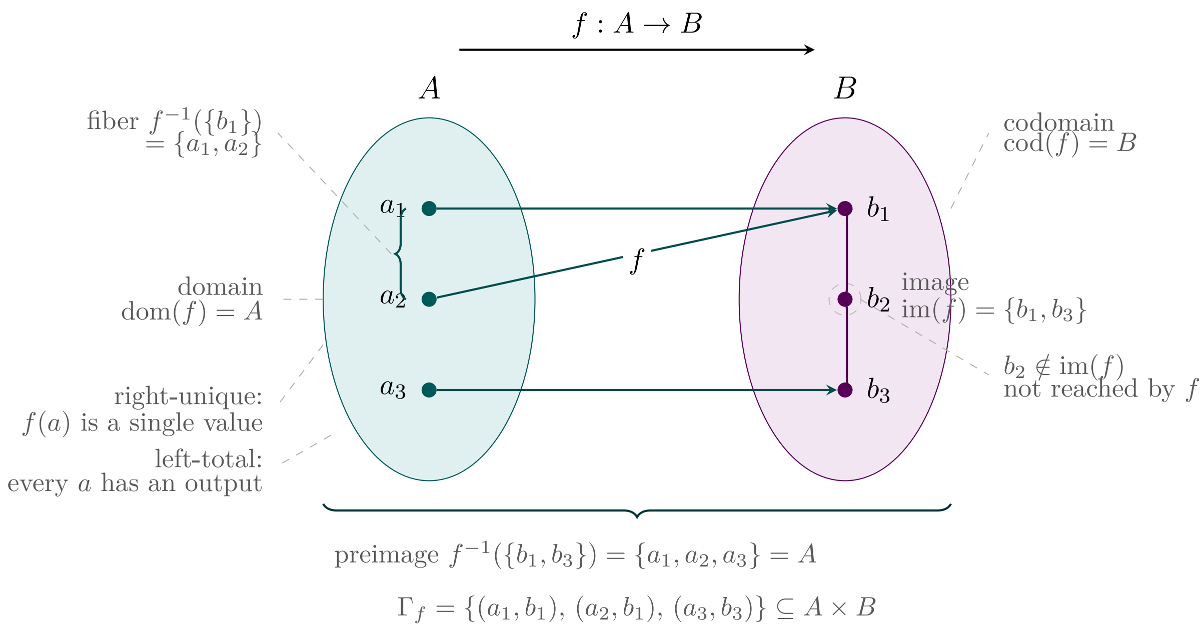


Figure 1.1: Vocabulary of a function $f : A \to B$.

Injective Functions

Definition (Injective)

Definition 1.5.2 (Injective Function)**.** Let $f : A \to B$. The function f is **injective** if

$$\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Remark (Standard quantified statement).

$$\text{Injective}(f, A, B) \iff f : A \to B \wedge \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Remark (Definition predicate reading). f is injective exactly when equal outputs force equal inputs.

Remark (Negated quantified statement).

$$\neg\,\text{Injective}(f, A, B) \iff \exists a_1, a_2 \in A,\ a_1 \neq a_2 \wedge f(a_1) = f(a_2).$$

Remark (Interpretation). Injectivity means distinct inputs do not collide under the function.

Remark (Dependencies).

- Function

Surjective Functions

Definition (Surjective)

Definition 1.5.3 (Surjective Function)**.** Let $f : A \to B$. The function f is **surjective** if

$$\forall b \in B,\ \exists a \in A,\ f(a) = b.$$

Remark (Standard quantified statement).

$$\text{Surjective}(f, A, B) \iff f : A \to B \land \forall b \in B,\ \exists a \in A,\ f(a) = b.$$

Remark (Definition predicate reading). f is surjective exactly when every element of the codomain is hit by some input.

Remark (Negated quantified statement).

$$\neg\,\text{Surjective}(f, A, B) \iff \exists b \in B,\ \forall a \in A,\ f(a) \neq b.$$

Remark (Interpretation). Surjectivity says that the image of f fills the whole codomain.

Remark (Dependencies).

- Function

Bijective Functions

Definition (Bijective)

Definition 1.5.4 (Bijective Function)**.** Let $f : A \to B$. The function f is **bijective** if it is both injective and surjective.

Remark (Standard quantified statement).

$$\text{Bijective}(f, A, B) \iff \text{Injective}(f, A, B) \land \text{Surjective}(f, A, B).$$

Remark (Definition predicate reading). f is bijective exactly when it is both injective and surjective.

Remark (Negated quantified statement).

$$\neg\,\text{Bijective}(f, A, B) \iff \neg\,\text{Injective}(f, A, B) \lor \neg\,\text{Surjective}(f, A, B).$$

Remark (Interpretation). Bijectivity is the set-theoretic part of invertibility. A bijective structure-preserving map is the usual route to an isomorphism.

Remark (Dependencies).

- Injective Function
- Surjective Function

Morphisms

Definition (Homomorphism)

Definition 1.5.5 (Homomorphism)**.** Let A and B be algebraic structures of the same type. A **homomorphism** $f : A \to B$ is a function that preserves the operations of the structure.

Remark (Standard quantified statement).

$$\text{Homomorphism}(f, A, B) \iff f : A \to B \text{ and } f \text{ preserves the named operations of } A \text{ and } B.$$

Remark (Definition predicate reading). f is a homomorphism exactly when it is a structure-preserving function.

Remark (Negated quantified statement).

$$\neg \operatorname{Homomorphism}(f, A, B)$$

when f is not a function $A \to B$ or fails to preserve at least one named operation.

Remark (Interpretation). Homomorphisms are the maps that keep algebraic operations visible through a function.

Remark (Dependencies).

- Function
- Binary Operation

Definition 1.5.6 (Monomorphism)**.** A **monomorphism** is an injective homomorphism.

Remark (Standard quantified statement).

$$\operatorname{Monomorphism}(f, A, B) \iff \operatorname{Homomorphism}(f, A, B) \wedge \operatorname{Injective}(f, A, B).$$

Remark (Definition predicate reading). A monomorphism is a homomorphism that embeds without collisions.

Remark (Negated quantified statement).

$$\neg \operatorname{Monomorphism}(f, A, B) \iff \neg \operatorname{Homomorphism}(f, A, B) \vee \neg \operatorname{Injective}(f, A, B).$$

Remark (Interpretation). Monomorphisms preserve structure while keeping distinct inputs distinct.

Remark (Dependencies).

- Homomorphism
- Injective Function

Definition 1.5.7 (Epimorphism)**.** An **epimorphism** is a surjective homomorphism.

Remark (Standard quantified statement).

$$\operatorname{Epimorphism}(f, A, B) \iff \operatorname{Homomorphism}(f, A, B) \wedge \operatorname{Surjective}(f, A, B).$$

Remark (Definition predicate reading). An epimorphism is a homomorphism that reaches every element of the codomain.

Remark (Negated quantified statement).

$$\neg \operatorname{Epimorphism}(f, A, B) \iff \neg \operatorname{Homomorphism}(f, A, B) \vee \neg \operatorname{Surjective}(f, A, B).$$

Remark (Interpretation). Epimorphisms preserve structure while covering the whole target.

Remark (Dependencies).

- Homomorphism
- Surjective Function

Definition 1.5.8 (Isomorphism)**.** An **isomorphism** is a bijective homomorphism.

Remark (Standard quantified statement).

$$\mathrm{Isomorphism}(f, A, B) \iff \mathrm{Homomorphism}(f, A, B) \wedge \mathrm{Bijective}(f, A, B).$$

Remark (Definition predicate reading). An isomorphism is a structure-preserving bijection.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Isomorphism}(f, A, B) \iff \neg\,\mathrm{Homomorphism}(f, A, B) \vee \neg\,\mathrm{Bijective}(f, A, B).$$

Remark (Interpretation). Isomorphisms identify two structures as the same up to relabeling.

Remark (Dependencies).

- Homomorphism
- Bijective Function

Definition 1.5.9 (Endomorphism)**.** An **endomorphism** is a homomorphism from a structure to itself.

Remark (Standard quantified statement).

$$\mathrm{Endomorphism}(f, A) \iff \mathrm{Homomorphism}(f, A, A).$$

Remark (Definition predicate reading). An endomorphism is a structure-preserving self-map.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Endomorphism}(f, A) \iff \neg\,\mathrm{Homomorphism}(f, A, A).$$

Remark (Interpretation). Endomorphisms are internal symmetries or transformations, not necessarily invertible ones.

Remark (Dependencies).

- Homomorphism

Definition 1.5.10 (Automorphism)**.** An **automorphism** is a bijective endomorphism.

Remark (Standard quantified statement).

$$\mathrm{Automorphism}(f, A) \iff \mathrm{Endomorphism}(f, A) \wedge \mathrm{Bijective}(f, A, A).$$

Remark (Definition predicate reading). An automorphism is an invertible structure-preserving self-map.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Automorphism}(f, A) \iff \neg\,\mathrm{Endomorphism}(f, A) \vee \neg\,\mathrm{Bijective}(f, A, A).$$

Remark (Interpretation). Automorphisms are the reversible symmetries of a structure.

Remark (Dependencies).

- Endomorphism
- Bijective Function

1.6 Distinguished Elements

Identity Elements

Definition 1.6.1 (Identity Element)**.** Let $\star$ be a binary operation on A. An element $e \in A$ is an **identity element** for $\star$ if

$$\forall a \in A, \quad e \star a = a \star e = a.$$

Remark (Standard quantified statement)*.*

$$\mathrm{IdentityElement}(e, \star, A) \iff e \in A \land \forall a \in A,\ e \star a = a \star e = a.$$

Remark (Definition predicate reading)*.* e is an identity for $\star$ exactly when multiplying by e on either side leaves every element unchanged.

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{IdentityElement}(e, \star, A) \iff e \notin A \lor \exists a \in A,\ e \star a \neq a \lor a \star e \neq a.$$

Remark (Interpretation)*.* An identity element is the distinguished element that does nothing under the operation.

Remark (Dependencies)*.*

- Binary Operation

Proposition 1.6.2 (Uniqueness of the Identity)**.** *In a group, the identity element is unique. Go to proof.*

Remark (Standard quantified statement)*.*

$$\mathrm{Group}(G, \star) \implies \exists! e \in G\ \ \mathrm{IdentityElement}(e, \star, G).$$

Remark (Theorem predicate reading)*.* Every group has exactly one identity element.

Remark (Negated quantified statement)*.*

$$\mathrm{Group}(G, \star) \land \exists e_1, e_2 \in G,\ \mathrm{IdentityElement}(e_1, \star, G) \land \mathrm{IdentityElement}(e_2, \star, G) \land e_1 \neq e_2 \implies \bot.$$

Remark (Interpretation)*.* The symbol for the identity is not extra arbitrary data once the group operation is fixed.

Remark (Dependencies)*.*

- Group
- Identity Element

Remark (Proof TODO)*.* Regenerate this proof as a one-proof-per-file artifact.

Inverses

Definition 1.6.3 (Inverse Element)**.** Let $\star$ be a binary operation on A with identity element e. An element $b \in A$ is an **inverse** of $a \in A$ if

$$a \star b = b \star a = e.$$

Remark (Standard quantified statement)*.*

$$\text{InverseElement}(b, a, \star, e, G) \iff a, b, e \in G \land a \star b = b \star a = e.$$

Remark (Definition predicate reading)*.* b is an inverse of a exactly when composing a and b in either order returns the identity.

Remark (Negated quantified statement)*.*

$$\neg\, \text{InverseElement}(b, a, \star, e, G) \iff b \notin G \lor a \star b \neq e \lor b \star a \neq e.$$

Remark (Interpretation)*.* An inverse is an element that undoes another element with respect to the group operation.

Remark (Dependencies)*.*

- Binary Operation
- Identity Element

Proposition 1.6.4 (Uniqueness of Inverses)**.** *In a group, every element has a unique inverse. Go to proof.*

Remark (Standard quantified statement)*.*

$$\text{Group}(G, \star) \implies \forall a \in G,\ \exists! b \in G\ \ \text{InverseElement}(b, a, \star, e, G).$$

Remark (Theorem predicate reading)*.* Every element of a group has exactly one inverse.

Remark (Negated quantified statement)*.*

$$\text{Group}(G, \star) \land \exists a, b, c \in G,\ \text{InverseElement}(b, a, \star, e, G) \land \text{InverseElement}(c, a, \star, e, G) \land b \neq c \implies \bot.$$

Remark (Interpretation)*.* Once the group operation and identity are fixed, inverse notation is well-defined.

Remark (Dependencies)*.*

- Group
- Inverse Element
- Identity Element

Remark (Proof TODO)*.* Regenerate this proof as a one-proof-per-file artifact.

Absorbing Elements

Definition 1.6.5 (Absorbing Element)**.** Let $\star$ be a binary operation on A. An element $z \in A$ is an **absorbing element** for $\star$ if

$$\forall a \in A, \quad z \star a = a \star z = z.$$

Remark (Standard quantified statement).

$$\text{AbsorbingElement}(z, \star, A) \iff z \in A \wedge \forall a \in A,\ z \star a = a \star z = z.$$

Remark (Definition predicate reading). z is absorbing for $\star$ exactly when combining any element with z on either side returns z.

Remark (Negated quantified statement).

$$\neg\,\text{AbsorbingElement}(z, \star, A) \iff z \notin A \vee \exists a \in A,\ z \star a \neq z \vee a \star z \neq z.$$

Remark (Interpretation). An absorbing element collapses every product involving it back to itself.

Remark (Examples). The element 0 is absorbing for multiplication in the integers, reals, and complex numbers.

Remark (Dependencies).

- Binary Operation

Distinguished Constants

Remark (Exposition). Distinguished constants are the model-theoretic presentation of named elements. The group identity e, additive identity 0, and multiplicative identity 1 are constants when they are part of the signature.

1.7 Algebraic Structures

Structure Signatures

Definition (First-Order Signature)

Definition 1.7.1 (First-Order Signature)**.** A **first-order signature** $\mathcal{L}$ consists of constant symbols, function symbols with specified arities, and relation symbols with specified arities.

Remark (Standard quantified statement).

$$\mathcal{L} = (\text{Const}_{\mathcal{L}}, \text{Func}_{\mathcal{L}}, \text{Rel}_{\mathcal{L}}, \text{arity})$$

where each function and relation symbol has a specified arity.

Remark (Definition predicate reading). A first-order signature is the syntactic list of symbols that a structure must interpret.

Remark (Interpretation). The signature describes what kinds of constants, operations, and relations are visible before any particular carrier set is chosen.

Remark (Dependencies).

- Nullary Operation
- Unary Operation
- Binary Operation
- Binary Relation

Definition ($\mathcal{L}$-Structure)

Definition 1.7.2 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** $\mathcal{A}$ consists of a nonempty carrier set A together with interpretations of each constant, function, and relation symbol of $\mathcal{L}$ on A.

Remark (Standard quantified statement).

$$\mathcal{A} \models \mathcal{L} \iff \mathcal{A} = (A, \ldots) \text{ interprets each symbol of } \mathcal{L} \text{ on } A.$$

Remark (Definition predicate reading). $\mathcal{A}$ is an $\mathcal{L}$-structure exactly when it gives meanings to the symbols of $\mathcal{L}$ on a carrier set.

Remark (Negated quantified statement).

$$\neg \, \mathrm{Structure}_{\mathcal{L}}(\mathcal{A})$$

when some symbol of $\mathcal{L}$ is not interpreted with the required arity on the carrier set.

Remark (Interpretation). The signature is the type; the structure is the instance. A group signature names multiplication, identity, and inverse, while a particular group interprets those symbols on a carrier set.

Remark (Dependencies).

- First-Order Signature
- Carrier Set

Magmas

Definition (Magma)

Definition 1.7.3 (Magma)**.** A **magma** is a pair $(A, \star)$ where A is a nonempty set and $\star : A \times A \to A$ is a binary operation.

Remark (Standard quantified statement).

$$\mathrm{Magma}(A, \star) \iff A \neq \varnothing \wedge \mathrm{BinaryOperation}(\star, A).$$

Remark (Definition predicate reading). $(A, \star)$ is a magma exactly when A is nonempty and $\star$ is closed on A.

Remark (Negated quantified statement).

$$\neg \, \mathrm{Magma}(A, \star) \iff A = \varnothing \vee \neg \, \mathrm{BinaryOperation}(\star, A).$$

Remark (Interpretation). A magma is the weakest nontrivial algebraic structure: it has a carrier set and a closed binary operation, with no associativity, identity, inverse, or commutativity requirements.

Remark (Dependencies).

- Carrier Set
- Binary Operation

Semigroups

Definition (Semigroup)

Definition 1.7.4 (Semigroup). A **semigroup** is a pair $(A, \star)$ where A is a nonempty set and $\star : A \times A \to A$ is an associative binary operation:

$$\forall a, b, c \in A, \quad (a \star b) \star c = a \star (b \star c).$$

Remark (Standard quantified statement).

$$\text{Semigroup}(A, \star) \iff \text{Magma}(A, \star) \wedge \text{Associative}(\star, A).$$

Remark (Definition predicate reading). $(A, \star)$ is a semigroup exactly when it is a magma whose operation is associative.

Remark (Negated quantified statement).

$$\neg\,\text{Semigroup}(A, \star) \iff \neg\,\text{Magma}(A, \star) \vee \neg\,\text{Associative}(\star, A).$$

Remark (Interpretation). Semigroups are magmas where finite products can be written without specifying parentheses.

Remark (Examples). Nonempty strings under concatenation and positive integers under addition are standard semigroups.

Remark (Dependencies).

- Magma
- Associativity

Monoids

Definition (Monoid)

Definition 1.7.5 (Monoid). A **monoid** is a triple $(A, \star, e)$ where $(A, \star)$ is a semigroup and $e \in A$ is an identity element:

$$\forall a \in A, \quad a \star e = e \star a = a.$$

Remark (Standard quantified statement).

$$\text{Monoid}(A, \star, e) \iff \text{Semigroup}(A, \star) \wedge \text{IdentityElement}(e, \star, A).$$

Remark (Definition predicate reading). $(A, \star, e)$ is a monoid exactly when $(A, \star)$ is a semigroup and e is a two-sided identity for $\star$.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Monoid}(A, \star, e) \iff \neg\,\mathrm{Semigroup}(A, \star) \vee \neg\,\mathrm{IdentityElement}(e, \star, A).$$

Remark (Interpretation). Monoids add a do-nothing element to associative composition.

Remark (Examples). The natural numbers under addition with identity 0, the natural numbers under multiplication with identity 1, and strings under concatenation with the empty string are monoids.

Remark (Dependencies).

- Semigroup
- Identity Element

Groups

Definition (Group)

Definition 1.7.6 (Group)**.** A **group** is a pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G satisfying:

G1. associativity;

G2. existence of an identity element $e \in G$;

G3. existence, for each $a \in G$, of an inverse element $a^{-1} \in G$.

Remark (Standard quantified statement).

$$\mathrm{Group}(G, \star) \iff \exists e \in G,\ \mathrm{Monoid}(G, \star, e) \wedge \forall a \in G,\ \exists b \in G\ \mathrm{InverseElement}(b, a, \star, e, G).$$

Remark (Definition predicate reading). $(G, \star)$ is a group exactly when it is an associative closed operation with an identity and an inverse for every element.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Group}(G, \star) \iff \neg\,\mathrm{Associative}(\star, G) \vee \text{no two-sided identity exists} \vee \text{some element has no two-sided in}$$

Remark (Interpretation). Groups are algebraic structures of reversible composition.

Remark (Dependencies).

- Binary Operation
- Associativity
- Identity Element
- Inverse Element

Definition 1.7.7 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its underlying set. If $|G|$ is finite, G is a finite group; otherwise it is an infinite group.

Remark (Standard quantified statement)*.*

$$\mathrm{OrderOfGroup}(G) = |G|.$$

Remark (Definition predicate reading)*.* The order of G is the number of elements in the carrier set of G.

Remark (Interpretation)*.* Finite groups are classified by a natural-number order; infinite groups have infinite cardinal order.

Remark (Examples)*.* Examples include $(\mathbb{Z}, +)$, $(\mathbb{Q}\setminus\{0\}, \cdot)$, $(\mathbb{Z}/n\mathbb{Z}, +)$, and $GL_n(\mathbb{R})$ under matrix multiplication.

Remark (Dependencies)*.*

- Group
- TODO: formalize cardinality.

Abelian Groups

Definition (Abelian Group)

Definition 1.7.8 (Abelian Group)**.** A group $(G, \star)$ is **abelian** if its operation is commutative:

$$\forall a, b \in G, \quad a \star b = b \star a.$$

Remark (Standard quantified statement)*.*

$$\mathrm{AbelianGroup}(G, \star) \iff \mathrm{Group}(G, \star) \wedge \mathrm{Commutative}(\star, G).$$

Remark (Definition predicate reading)*.* $(G, \star)$ is abelian exactly when it is a group whose operation commutes.

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{AbelianGroup}(G, \star) \iff \neg\,\mathrm{Group}(G, \star) \vee \exists a, b \in G,\ a \star b \neq b \star a.$$

Remark (Interpretation)*.* Abelian groups are usually written additively: the operation is $+$, the identity is 0, and the inverse of a is $-a$.

Remark (Dependencies)*.*

- Group
- Commutativity

Rings

Definition (Ring)

Definition 1.7.9 (Ring)**.** A **ring** is a triple $(R, +, \cdot)$ where R is a set and $+$ and $\cdot$ are binary operations on R satisfying:

R1. $(R, +)$ is an abelian group;

R2. multiplication is associative;

R3. multiplication distributes over addition on the left and on the right.

Remark (Standard quantified statement).

$$\mathrm{Ring}(R,+,\cdot) \iff \mathrm{AbelianGroup}(R,+) \wedge \mathrm{Associative}(\cdot, R) \wedge \mathrm{LeftDistributive}(\cdot,+,R) \wedge \mathrm{RightDistributive}($$

Remark (Definition predicate reading). $(R, +, \cdot)$ is a ring exactly when addition forms an abelian group, multiplication is associative, and multiplication distributes over addition.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Ring}(R,+,\cdot) \iff \neg\,\mathrm{AbelianGroup}(R,+) \vee \neg\,\mathrm{Associative}(\cdot, R) \vee \neg\,\mathrm{LeftDistributive}(\cdot,+,R) \vee \neg\,\mathrm{RightDis}$$

Remark (Interpretation). A ring is the algebraic setting where addition, subtraction, and multiplication coexist coherently.

Remark (Dependencies).

- Abelian Group
- Associativity
- Left Distributivity
- Right Distributivity

Definition 1.7.10 (Commutative Ring)**.** A ring $(R, +, \cdot)$ is **commutative** if

$$\forall a, b \in R, \quad a \cdot b = b \cdot a.$$

Remark (Standard quantified statement).

$$\mathrm{CommutativeRing}(R,+,\cdot) \iff \mathrm{Ring}(R,+,\cdot) \wedge \mathrm{Commutative}(\cdot, R).$$

Remark (Definition predicate reading). A ring is commutative exactly when its multiplication commutes.

Remark (Negated quantified statement).

$$\neg\,\mathrm{CommutativeRing}(R,+,\cdot) \iff \neg\,\mathrm{Ring}(R,+,\cdot) \vee \exists a, b \in R,\ a \cdot b \neq b \cdot a.$$

Remark (Interpretation). Commutative rings are rings whose multiplication behaves symmetrically in its two inputs.

Remark (Dependencies).

- Ring
- Commutativity

Definition 1.7.11 (Ring with Unity)**.** A ring $(R, +, \cdot)$ is a **ring with unity** if there exists $1 \in R$ such that

$$\forall a \in R, \quad 1 \cdot a = a \cdot 1 = a.$$

Remark (Standard quantified statement).

$$\mathrm{RingWithUnity}(R, +, \cdot, 1) \iff \mathrm{Ring}(R, +, \cdot) \wedge \mathrm{IdentityElement}(1, \cdot, R).$$

Remark (Definition predicate reading). A ring has unity exactly when multiplication has a two-sided identity.

Remark (Negated quantified statement).

$$\neg\,\mathrm{RingWithUnity}(R, +, \cdot, 1) \iff \neg\,\mathrm{Ring}(R, +, \cdot) \vee \neg\,\mathrm{IdentityElement}(1, \cdot, R).$$

Remark (Interpretation). The unity element is the multiplicative identity of the ring.

Remark (Examples). The integers, residue rings $\mathbb{Z}/n\mathbb{Z}$, polynomial rings, and matrix rings are standard examples of rings.

Remark (Dependencies).

- Ring
- Identity Element

Integral Domains

Definition (Zero Divisor)**.** Let R be a ring. A nonzero element $a \in R$ is a **zero divisor** if there exists a nonzero element $b \in R$ such that $ab = 0$.

Remark (Canonical reference). The canonical dependency label for this vocabulary is Zero Divisor in Volume II.

Remark (Standard quantified statement).

$$\mathrm{ZeroDivisor}(a, R) \iff a \in R \wedge a \neq 0 \wedge \exists b \in R,\ b \neq 0 \wedge ab = 0.$$

Remark (Definition predicate reading). a is a zero divisor exactly when a nonzero multiplication by a can produce zero.

Remark (Negated quantified statement).

$$\neg\,\mathrm{ZeroDivisor}(a, R) \iff a = 0 \vee \forall b \in R,\ ab = 0 \Rightarrow b = 0.$$

Remark (Interpretation). Zero divisors identify where multiplication loses cancellation behavior.

Remark (Dependencies).

- Ring

- Absorbing Element

Definition (Integral Domain)

Definition 1.7.12 (Integral Domain)**.** An **integral domain** is a commutative ring with unity in which there are no zero divisors.

Remark (Standard quantified statement).

$$\mathrm{IntegralDomain}(R,+,\cdot,0,1) \iff \mathrm{CommutativeRing}(R,+,\cdot) \wedge \mathrm{RingWithUnity}(R,+,\cdot,1) \wedge \forall a,b \in R,\ ab$$

Remark (Definition predicate reading). An integral domain is a commutative ring with 1 where a product is zero only because at least one factor is zero.

Remark (Negated quantified statement).

$$\neg\,\mathrm{IntegralDomain}(R,+,\cdot,0,1) \iff \neg\,\mathrm{CommutativeRing}(R,+,\cdot) \vee \neg\,\mathrm{RingWithUnity}(R,+,\cdot,1) \vee \exists a,b \in$$

Remark (Interpretation). Integral domains are the commutative rings where multiplication has no nonzero zero-divisor obstruction.

Remark (Dependencies).

- Commutative Ring
- Ring with Unity
- Zero Divisor

Proposition 1.7.13 (Cancellation in Integral Domains)**.** *Let R be an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$. Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{IntegralDomain}(R) \implies \forall a,b,c \in R,\ a \neq 0 \wedge ab = ac \Rightarrow b = c.$$

Remark (Theorem predicate reading). Nonzero factors can be cancelled in an integral domain.

Remark (Negated quantified statement).

$$\mathrm{IntegralDomain}(R) \wedge \exists a,b,c \in R,\ a \neq 0 \wedge ab = ac \wedge b \neq c \implies \perp.$$

Remark (Interpretation). Cancellation is recovered in integral domains because nonzero elements are not zero divisors.

Remark (Dependencies).

- Integral Domain
- Zero Divisor

Remark (Proof TODO). Regenerate this proof as a one-proof-per-file artifact.

Fields

Definition (Field)

Definition 1.7.14 (Field)**.** A **field** is a set F equipped with two binary operations, addition $+$ and multiplication $\cdot$, such that:

F1. $(F,+)$ is an abelian group with identity 0;

F2. $F \setminus \{0\}$ is an abelian group under multiplication with identity 1;

F3. multiplication distributes over addition.

Remark (Standard quantified statement).

$$\mathrm{Field}(F,+,\cdot,0,1) \iff \mathrm{AbelianGroup}(F,+) \wedge \mathrm{AbelianGroup}(F \setminus \{0\},\cdot) \wedge \mathrm{LeftDistributive}(\cdot,+,F) \wedge \mathrm{RightI}$$

Remark (Definition predicate reading). A field is an additive abelian group whose nonzero elements form a multiplicative abelian group, with multiplication distributing over addition.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Field}(F,+,\cdot,0,1) \iff \neg\,\mathrm{AbelianGroup}(F,+) \vee \neg\,\mathrm{AbelianGroup}(F \setminus \{0\},\cdot) \vee \neg\,\mathrm{LeftDistributive}(\cdot,+,F)$$

Remark (Interpretation). Fields are the algebraic setting where addition, subtraction, multiplication, and division by nonzero elements are all available.

Remark (Dependencies).

- Abelian Group
- Left Distributivity
- Right Distributivity

Proposition 1.7.15 (Every Field is an Integral Domain)**.** *Every field is an integral domain. Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{Field}(F,+,\cdot,0,1) \implies \mathrm{IntegralDomain}(F,+,\cdot,0,1).$$

Remark (Theorem predicate reading). The field axioms imply the integral-domain axioms.

Remark (Negated quantified statement).

$$\mathrm{Field}(F,+,\cdot,0,1) \wedge \neg\,\mathrm{IntegralDomain}(F,+,\cdot,0,1) \implies \bot.$$

Remark (Interpretation). The existence of inverses for nonzero elements prevents nonzero zero divisors.

Remark (Dependencies).

- Field
- Integral Domain
- Inverse Element

Remark (Proof TODO). Regenerate this proof as a one-proof-per-file artifact.

1.8 Groups

Definition (Binary Operation)

Definition 1.8.1 (Binary Operation)**.** Let G be a set. A **binary operatio** is a function

$$\star : G \times G \to G.$$

For $a, b \in G$, the element $\star(a, b)$ is denoted by $a \star b$.

Binary Operations

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\star \text{ is a binary operation on } G \\ \iff\quad &\forall a \in G\ \forall b \in G,\ a \star b \in G. \end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{BinaryOperation}(\star, G) \;:\equiv\; \forall a \in G\ \forall b \in G,\ a \star b \in G.$$

Remark (Negated quantified statement)*.*

$$\begin{aligned} &\star \text{ is not a binary operation on } G \\ \iff\quad &\exists a \in G\ \exists b \in G \text{ such that } a \star b \notin G. \end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{BinaryOperation}(\star, G) \;:\equiv\; \exists a \in G\ \exists b \in G \text{ such that } a \star b \notin G.$$

Remark (Failure modes)*.*

$$\exists a \in G\ \exists b \in G \text{ such that } a \star b \notin G.$$

Remark (Failure mode decomposition)*.*

$$\begin{aligned} &\star \text{ fails to be a binary operation on } G \\ \iff\quad &\exists a \in G\ \exists b \in G \text{ such that } a \star b \notin G. \end{aligned}$$

Remark (Interpretation)*.* A binary operation takes two elements of G and returns another element of G. This encodes closure of the operation within the set.

Definition (Associativity)

Definition 1.8.2 (Associativity)**.** Let $\star$ be a binary operation on G. The operation $\star$ is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in G.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\star \text{ is associative on } G \\ \iff\quad &\forall a \in G\ \forall b \in G\ \forall c \in G,\ (a \star b) \star c = a \star (b \star c). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{Associative}(\star, G) \;:\equiv\; \forall a \in G\ \forall b \in G\ \forall c \in G,\ (a \star b) \star c = a \star (b \star c).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\star \text{ is not associative on } G \\ \iff\quad &\exists a \in G\ \exists b \in G\ \exists c \in G \text{ such that} \\ &(a \star b) \star c \neq a \star (b \star c). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{Associative}(\star, G) \;:\equiv\; \exists a \in G\ \exists b \in G\ \exists c \in G \text{ such that } (a \star b) \star c \neq a \star (b \star c).$$

Remark (Failure modes).

$$\exists a \in G\ \exists b \in G\ \exists c \in G \text{ such that } (a \star b) \star c \neq a \star (b \star c).$$

Remark (Failure mode decomposition).

$$\begin{aligned} &\star \text{ fails associativity} \\ \iff\quad &\exists a, b, c \in G \text{ witnessing a parenthesization mismatch.} \end{aligned}$$

Remark (Interpretation). Associativity means that the way elements are grouped does not affect the result.

Definition (Commutativity)

Definition 1.8.3 (Commutativity)**.** Let $\star$ be a binary operation on G. The operation $\star$ is **commutative** if

$$a \star b = b \star a \quad \text{for all } a, b \in G.$$

Remark (Standard quantified statement).

$$\begin{aligned} &\star \text{ is commutative on } G \\ \iff\quad &\forall a \in G\ \forall b \in G,\ a \star b = b \star a. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{Commutative}(\star, G) \;:\equiv\; \forall a \in G\ \forall b \in G,\ a \star b = b \star a.$$

Remark (Negated quantified statement).

$$\begin{aligned} &\star \text{ is not commutative on } G \\ \iff\quad &\exists a \in G\ \exists b \in G \text{ such that } a \star b \neq b \star a. \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{Commutative}(\star, G) \;:\equiv\; \exists a \in G\ \exists b \in G \text{ such that } a \star b \neq b \star a.$$

Remark (Failure modes).

$$\exists a \in G\ \exists b \in G \text{ such that } a \star b \neq b \star a.$$

Remark (Failure mode decomposition).

$$\begin{aligned} &\star \text{ fails commutativity} \\ &\iff \quad \exists a, b \in G \text{ where swapping inputs changes the result.} \end{aligned}$$

Remark (Interpretation). Commutativity means the order of the inputs does not affect the result.

Remark (Interpretation). Associativity and commutativity are independent properties. For example, addition on $\mathbb{Z}$ is both associative and commutative; matrix multiplication is associative but not commutative; subtraction on $\mathbb{Z}$ is neither.

Definition of a Group

Definition 1.8.4 (Group)**.** A *group* is a pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G satisfying the following axioms:

G1. Associativity. $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$.

G2. Identity. There exists an element $e \in G$ such that

$$e \star a = a \star e = a \quad \text{for all } a \in G.$$

The element e is called the *identity element* of G.

G3. Inverses. For each $a \in G$, there exists an element $a^{-1} \in G$ such that

$$a \star a^{-1} = a^{-1} \star a = e.$$

The element a^{-1} is called the *inverse* of a.

Remark 1.8.5. Closure is not listed as a separate axiom because it is already encoded in the requirement that $\star : G \times G \to G$ is a binary operation — the codomain forces the result to stay in G.

Intuitively: a group is a set where you can combine elements, undo combinations, and the order of grouping never matters.

Remark 1.8.6 (Axiom Count). Some treatments list four axioms (closure, associativity, identity, inverses). Here closure is absorbed into the definition of binary operation, leaving three axioms. Both presentations define the same object.

Remark 1.8.7 (Notation). When the operation is understood from context, we write ab instead of $a \star b$, and call G itself a group rather than the pair $(G, \star)$. For groups whose operation is addition, we write $a + b$, use 0 for the identity, and $-a$ for the inverse of a.

Definition 1.8.8 (Order of a Group)**.** The *order* of a group G, denoted $|G|$, is the cardinality of G as a set. If $|G|$ is finite, G is called a *finite group*; otherwise it is an *infinite group*.

Example 1.8.9 (Canonical Examples of Groups).

(i) $(\mathbb{Z}, +)$: the integers under addition. Identity: 0. Inverse of n: $-n$. Infinite group.

(ii) $(\mathbb{Q} \setminus \{0\}, \cdot)$: nonzero rationals under multiplication. Identity: 1. Inverse of q: $1/q$. Infinite group.

(iii) $(\mathbb{Z}/n\mathbb{Z}, +)$: integers modulo n under addition. Identity: $[0]$. Inverse of $[k]$: $[n-k]$. Finite group of order n.

(iv) $(GL_n(\mathbb{R}), \cdot)$: invertible $n \times n$ real matrices under multiplication. Identity: I_n. Inverse: matrix inverse. Infinite group.

Remark 1.8.10. Note what fails to be a group: $(\mathbb{Z}, \cdot)$ is not a group because 2 has no multiplicative inverse in $\mathbb{Z}$. $(\mathbb{N}, +)$ is not a group because positive integers have no additive inverse in $\mathbb{N}$. These failures illustrate why each axiom is necessary.

Basic Theorems

Remark 1.8.11 (Why These Theorems Matter). The group axioms guarantee existence of an identity and inverses, but say nothing about uniqueness. The following theorems establish that both are unique. This is essential: without uniqueness, we cannot speak of *the* identity or *the* inverse of an element, and proofs that equate two objects via the identity or inverse would be invalid.

These same uniqueness theorems reappear in vector space proofs, where they are cited by name. They are proved once here.

Proposition 1.8.12 (Uniqueness of the Identity)**.** *Let G be a group. The identity element of G is unique.* *Go to proof.*

Remark.

Remark 1.8.13. The proof strategy is standard for uniqueness arguments: assume two identities exist, then show they must be equal. This pattern recurs throughout algebra whenever a definition asserts existence of a distinguished element.

Intuitively: if two elements both act as an identity, applying one to the other forces them to coincide.

Proposition 1.8.14 (Uniqueness of Inverses)**.** *Let G be a group. For each $a \in G$, the inverse of a is unique.* *Go to proof.*

Remark.

Remark 1.8.15. Intuitively: if two elements both undo a, then they must be the same element — because each can be obtained from the other by cancellation.

Proposition 1.8.16 (Cancellation Laws)**.** *Let G be a group and let $a, b, c \in G$. Then:*

(i) ***Left cancellation:*** $ab = ac \implies b = c$.

(ii) ***Right cancellation:*** $ba = ca \implies b = c$.

Go to proof.

Remark 1.8.17. Cancellation is what makes group equations solvable. It does *not* hold in general for rings or monoids without inverses.

Intuitively: multiply both sides by a^{-1} and the common factor disappears.

Proposition 1.8.18 (Socks-Shoes Property)**.** *Let G be a group and let $a, b \in G$. Then*

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Go to proof.

Remark.

Remark 1.8.19. The name comes from the observation that to undo putting on socks then shoes, you must first remove the shoes, then the socks — in reverse order.

This reversal of order is characteristic of non-abelian groups and becomes important in the theory of group homomorphisms and in matrix algebra, where $(AB)^{-1} = B^{-1}A^{-1}$.

Abelian Groups

Definition 1.8.20 (Abelian Group)**.** A group $(G, \star)$ is called *abelian* (or *commutative*) if

$$a \star b = b \star a \quad \text{for all } a, b \in G.$$

Remark 1.8.21. An abelian group is a group with one additional axiom: commutativity of the operation. Every abelian group is a group, but not every group is abelian.

Intuitively: in an abelian group, the order in which you combine elements is irrelevant. This is the familiar arithmetic of addition on $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.

Remark 1.8.22 (Structural Position)*.* Abelian groups are the additive backbone of every vector space. The four vector space axioms governing addition — associativity, commutativity, existence of zero, existence of additive inverses — are precisely the axioms that make $(V, +)$ an abelian group.

This is why the vector space definition can be stated compactly as: *a vector space over* $\mathbb{F}$ *is an abelian group* $(V, +)$ *equipped with a scalar multiplication by* $\mathbb{F}$. The abelian group structure is not an analogy; it is the literal algebraic content of the first four vector space axioms.

Example 1.8.23 (Abelian and Non-Abelian Groups)*.*

(i) $(\mathbb{Z}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$: all abelian. These are the additive groups underlying the standard vector spaces.

(ii) $(\mathbb{Z}/n\mathbb{Z}, +)$: abelian for all $n \geq 1$.

(iii) $(GL_n(\mathbb{R}), \cdot)$ for $n \geq 2$: *not* abelian, since matrix multiplication does not commute in general.

(iv) $(S_n, \circ)$ for $n \geq 3$: the symmetric group on n symbols under composition is not abelian.

Remark 1.8.24 (Additive Notation Convention)*.* For abelian groups, it is standard to write the operation as $+$, the identity as 0, and the inverse of a as $-a$. This additive notation is used throughout linear algebra, where $(V, +)$ is always an abelian group.

1.9 Rings

Definition of a Ring

Remark 1.9.1 (Motivation)*.* A group has one binary operation. A ring has two: addition and multiplication. The addition makes the ring an abelian group. Multiplication is layered on top, connected to addition through the distributive laws.

Rings are the natural algebraic home of arithmetic. The integers $\mathbb{Z}$, polynomials $\mathbb{Z}[x]$, and square matrices $M_n(\mathbb{R})$ are all rings. What they share is not the specific objects but the axiom structure — and every theorem proved here holds simultaneously in all of them.

Definition 1.9.2 (Ring)**.** A *ring* is a triple $(R, +, \cdot)$ where R is a set and $+$ and $\cdot$ are binary operations on R satisfying:

R1. Additive abelian group. $(R, +)$ is an abelian group:

- $(a + b) + c = a + (b + c)$ for all $a, b, c \in R$.
- $a + b = b + a$ for all $a, b \in R$.
- There exists $0 \in R$ such that $a + 0 = a$ for all $a \in R$.
- For each $a \in R$, there exists $-a \in R$ such that $a + (-a) = 0$.

R2. Multiplicative associativity. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$.

R3. Distributivity.

- $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in R$.
- $(a + b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.

Remark 1.9.3 (What a Ring Does and Does Not Require). A ring does *not* require:

- commutativity of multiplication ($ab = ba$ need not hold),
- a multiplicative identity (1 need not exist),
- multiplicative inverses (a^{-1} need not exist).

Each additional requirement produces a richer structure. A ring with a multiplicative identity is a *ring with unity.* A commutative ring with unity where every nonzero element has a multiplicative inverse is a *field* — covered in the next section.

Intuitively: a ring is the minimal structure needed for addition, subtraction, and multiplication to coexist and interact sensibly. Division is not guaranteed.

Definition 1.9.4 (Commutative Ring)**.** A ring $(R, +, \cdot)$ is *commutative* if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Definition 1.9.5 (Ring with Unity)**.** A ring $(R, +, \cdot)$ is a *ring with unity* if there exists $1 \in R$ such that

$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$

The element 1 is called the *multiplicative identity* or *unity.* When it exists, it is unique (proof identical to Proposition 1.8.12 applied to $(R, \cdot)$).

Example 1.9.6 (Canonical Examples of Rings).

(i) $(\mathbb{Z}, +, \cdot)$: integers. Commutative ring with unity 1. No multiplicative inverses for $|n| \neq 1$.

(ii) $(\mathbb{Z}/n\mathbb{Z}, +, \cdot)$: integers modulo n. Commutative ring with unity $[1]$.

(iii) $(M_n(\mathbb{R}), +, \cdot)$: $n \times n$ real matrices. Ring with unity I_n. *Not* commutative for $n \geq 2$.

(iv) $(\mathbb{Z}[x], +, \cdot)$: polynomials with integer coefficients. Commutative ring with unity 1.

(v) The trivial ring $\{0\}$, where $0 = 1$, is the only ring in which the additive and multiplicative identities coincide.

Basic Theorems

Remark 1.9.7 (What Needs Proving). The ring axioms say nothing explicitly about how multiplication interacts with the additive identity 0 or with additive inverses. These must be derived. The key tool in both proofs is distributivity — the bridge between the two operations.

Notice that these proofs make no assumption about which ring you are in. They work in $\mathbb{Z}$, in $M_n(\mathbb{R})$, in $\mathbb{Z}[x]$, in any ring simultaneously.

Proposition 1.9.8 (Multiplication by Zero)**.** *Let R be a ring. For all $a \in R$,*

$$a \cdot 0 = 0 \cdot a = 0.$$

Go to proof.

Remark.

Remark 1.9.9. The proof uses distributivity to produce the equation $a \cdot 0 + a \cdot 0 = a \cdot 0$, then applies additive cancellation to conclude $a \cdot 0 = 0$.

This is not circular: 0 on the left side is the additive identity of the ring; the 0 on the right is the same element being derived as a consequence of cancellation. The proof works because additive cancellation was already proved for all abelian groups (Proposition 1.4.6).

Proposition 1.9.10 (Multiplication by Additive Inverse)**.** *Let R be a ring. For all $a, b \in R$,*

$$a \cdot (-b) = -(a \cdot b) \quad \textit{and} \quad (-a) \cdot b = -(a \cdot b).$$

In any ring with unity, $(-1) \cdot a = -a$. *Go to proof.*

Remark.

Remark 1.9.11. The proof strategy is: show $a \cdot (-b)$ satisfies the defining property of the additive inverse of $a \cdot b$, then invoke uniqueness of additive inverses (Proposition 1.8.14).

This is the same strategy used in the vector space proof that $(-1)v = -v$ — because that proof is just this theorem applied to the scalar field $\mathbb{F}$ acting on V.

Integral Domains

Remark 1.9.12 (Motivation). In $\mathbb{Z}$, if $ab = 0$ then $a = 0$ or $b = 0$. This feels obvious but it is not an axiom of rings — it fails in $\mathbb{Z}/6\mathbb{Z}$, where $[2] \cdot [3] = [0]$ even though neither $[2]$ nor $[3]$ is zero. Elements that behave like $[2]$ and $[3]$ are called zero divisors, and rings without them are integral domains.

This property matters because it is exactly what is needed for multiplicative cancellation — and for the zero product argument that appears throughout linear algebra.

Definition 1.9.13 (Zero Divisor)**.** Let R be a commutative ring with unity. A nonzero element $a \in R$ is a *zero divisor* if there exists a nonzero $b \in R$ such that $a \cdot b = 0$.

Definition 1.9.14 (Integral Domain)**.** A commutative ring with unity R is an *integral domain* if R has no zero divisors. Equivalently,

$$a \cdot b = 0 \implies a = 0 \;\text{ or }\; b = 0 \quad \text{for all } a, b \in R.$$

Remark 1.9.15 (Connection to Linear Algebra). The zero product property is the theorem cited in the vector space proof that $av = \mathbf{0} \Rightarrow a = 0$ or $v = \mathbf{0}$. Specifically, the case $a \neq 0$ uses the fact that the scalar field $\mathbb{F}$ has no zero divisors — which holds because every field is an integral domain (proved in the next section).

The proof is not about vectors at all. It is about the scalar field.

Proposition 1.9.16 (Cancellation in Integral Domains). *Let R be an integral domain and $a, b, c \in R$ with $a \neq 0$. Then*

$$ab = ac \implies b = c.$$

Go to proof.

Remark.

Example 1.9.17 (Integral Domains and Non-Examples).

(i) $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$: all integral domains.

(ii) $\mathbb{Z}[x]$: integral domain.

(iii) $\mathbb{Z}/p\mathbb{Z}$ for prime p: integral domain (in fact a field, as shown in the next section).

(iv) $\mathbb{Z}/6\mathbb{Z}$: *not* an integral domain, since $[2][3] = [0]$ with $[2], [3] \neq [0]$.

(v) $M_2(\mathbb{R})$: *not* an integral domain — not commutative, and admits zero divisors.

1.10 Fields

Definition of a Field

Remark 1.10.1 (Motivation). A ring allows addition, subtraction, and multiplication. A field adds division: every nonzero element has a multiplicative inverse. This is the structure of $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ — the number systems where you can always solve $ax = b$ for $a \neq 0$.

Fields are the scalars of linear algebra. Axler's $\mathbb{F}$ denotes an arbitrary field, meaning every theorem in linear algebra holds simultaneously over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$, and any other field — because the proofs use only field axioms, not properties specific to real or complex numbers.

Definition 1.10.2 (Field). A *field* is a commutative ring with unity $(F, +, \cdot)$ satisfying:

F1. Additive abelian group. $(F, +)$ is an abelian group with identity 0.

F2. Multiplicative abelian group on nonzero elements. $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1. Explicitly: for each $a \in F$ with $a \neq 0$, there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.

F3. Distributivity. $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in F$.

F4. Non-triviality. $0 \neq 1$.

Remark 1.10.3 (Unpacking the Definition). A field is simultaneously:

- $(F, +)$: an abelian group (additive structure),

- $(F \setminus \{0\}, \cdot)$: an abelian group (multiplicative structure),
- connected by distributivity.

The non-triviality axiom $0 \neq 1$ rules out the trivial ring $\{0\}$, which would otherwise technically satisfy the other axioms.

Example 1.10.4 (Fields). (i) $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$: the standard fields.

(ii) $\mathbb{Z}/p\mathbb{Z}$ for any prime p: a finite field with p elements, denoted $\mathbb{F}_p$.

(iii) $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$: a field extending $\mathbb{Q}$.

Example 1.10.5 (Non-Fields). (i) $\mathbb{Z}$: not a field. The element 2 has no multiplicative inverse in $\mathbb{Z}$.

(ii) $\mathbb{Z}/6\mathbb{Z}$: not a field. $[2]$ has no inverse since $\gcd(2, 6) \neq 1$. (Also not an integral domain.)

(iii) $M_n(\mathbb{R})$ for $n \geq 2$: not a field. Not commutative, and singular matrices have no inverse.

Basic Theorems

Remark 1.10.6 (Why These Theorems Matter). These are the theorems cited in every vector space proof that crosses from the scalar side to the vector side. They live here, in the field, and are cited by number when needed in linear algebra.

Proposition 1.10.7 (Every Field is an Integral Domain)**.** *Every field is an integral domain. Go to proof.*

Remark.

Remark 1.10.8. This is the theorem behind the linear algebra argument: if $av = \mathbf{0}$ and $a \neq 0$, then $v = \mathbf{0}$. The scalar a lives in a field $\mathbb{F}$, and the key step is that $\mathbb{F}$ has no zero divisors.

Proposition 1.10.9 (Zero Product Property in a Field)**.** *Let $\mathbb{F}$ be a field and $a, b \in \mathbb{F}$. Then*

$$ab = 0 \implies a = 0 \quad \text{or} \quad b = 0.$$

Go to proof.

Remark.

Proposition 1.10.10 (Nonzero Scalars Have Inverses)**.** *Let $\mathbb{F}$ be a field and $a \in \mathbb{F}$ with $a \neq 0$. Then there exists a unique $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$. Go to proof.*

Remark.

Remark 1.10.11. This is the theorem cited in the vector space proof that $av = \mathbf{0}$ with $a \neq 0$ implies $v = \mathbf{0}$: the step "multiply both sides by a^{-1}" is valid precisely because a^{-1} exists and is unique.

Proposition 1.10.12 (Characteristic of a Field)**.** *Let $\mathbb{F}$ be a field. The* characteristic *of $\mathbb{F}$ is the smallest positive integer n such that*

$$\underbrace{1 + 1 + \cdots + 1}_{n} = 0,$$

or 0 if no such n exists. The characteristic of a field is either 0 or a prime p. Go to proof.

Remark.

Remark 1.10.13. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0. $\mathbb{Z}/p\mathbb{Z}$ has characteristic p.

In Axler, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$, both of characteristic 0. Some results (such as the existence of eigenvalues) require characteristic 0 or algebraic closure and would fail over $\mathbb{F}_p$.

Bridge to Linear Algebra

Remark 1.10.14 (Why This Chapter Exists). Every proof in Axler's Chapter 1 draws on exactly two sources: properties of the field $\mathbb{F}$, and properties of the vector space V. This chapter is the permanent home of the field side. The table below maps each type of linear algebra proof step to the theorem it cites from this chapter.

Proof step in linear algebra	**Structure used**	**Theorem**
"since $a \neq 0$, a^{-1} exists"	Field $\mathbb{F}$	Prop. 1.10.10
"$ab = 0$ and $a \neq 0$ implies $b = 0$"	Field $\mathbb{F}$	Prop. 1.10.9
"$(-1)v = -v$"	Ring theorem on $\mathbb{F}$	Prop. 1.9.10
"$0 \cdot v = \mathbf{0}$"	Ring theorem on $\mathbb{F}$ acting on V	Prop. 1.9.8
"the additive identity is unique"	Abelian group $(V, +)$	Prop. 1.8.12
"additive inverses are unique"	Abelian group $(V, +)$	Prop. 1.8.14
"$-(-v) = v$"	Abelian group $(V, +)$	Prop. 1.8.14

Remark 1.10.15 (The Interaction Layer). The table above separates into two kinds of steps: those that use the structure of the field $\mathbb{F}$ alone (rows 1–3), and those that use the abelian group structure of V alone (rows 5–7). Row 4 is the interaction: it uses a field theorem applied across scalar multiplication to a vector.

This separation is the heart of the vector space structure. The field $\mathbb{F}$ and the abelian group V are independent objects connected by the scalar multiplication axioms. When a proof crosses that connection, it is using the interaction layer. When it stays on one side, it is using either field theory or group theory alone. Knowing which side you are on at each step is what the DU/TA/AM annotation system makes explicit.

1.11 Number Systems as Structures

The Arithmetic Signature

Definition 1.11.1 (Arithmetic Signature)**.** The **arithmetic signature** is the collection of symbols used to view number systems as structures: arithmetic operations such as $+$ and $\cdot$, distinguished constants such as 0 and 1, and relations such as $\leq$ when order is included.

Remark (Standard quantified statement)*.*

$$\mathcal{L}_{\text{arith}} = \{+, \cdot, 0, 1, \leq, \dots\},$$

with the visible symbols chosen according to the intended number-system structure.

Remark (Definition predicate reading)*.* The arithmetic signature is the symbol list used before a particular number system interprets those symbols.

Remark (Interpretation)*.* Different number systems interpret the same visible arithmetic symbols on different carrier sets. For example, addition on $\mathbb{N}$ and addition on $\mathbb{R}$ have the same symbol but different domains and different structural consequences.

Remark (Dependencies)*.*

- First-Order Signature
- Unary Operation
- Binary Operation
- Nullary Operation

The Natural Numbers as a Structure

Remark (Modeling Goal)*.* We exhibit $\mathbb{N}$ as a concrete *model*: fix the arithmetic signature, take the carrier and operations built in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{N}$ obeys the relevant natural-number axioms. The successor fragment recovers the Peano-system view; the additive and multiplicative fragments are obtained by reduction.

Remark (Exposition)*.* A model is a signature together with an interpretation on a carrier. The signature here is the natural-number fragment of the arithmetic signature,

$$\mathcal{L}_{\mathbb{N}} = \{S, +, \cdot, 0, 1\},$$

with unary function symbol S, binary function symbols $+, \cdot$, and constant symbols $0, 1$. The order symbol $<$ is deliberately absent; it enters by expansion below.

Definition 1.11.2 (The Natural Number Model $\mathcal{N}$)**.** The **natural number model** is the $\mathcal{L}_{\mathbb{N}}$-structure

$$\mathcal{N} = \big(\mathbb{N};\ S^{\mathcal{N}},\ +^{\mathcal{N}},\ \cdot^{\mathcal{N}},\ 0^{\mathcal{N}},\ 1^{\mathcal{N}}\big),$$

whose carrier $|\mathcal{N}| = \mathbb{N}$ is the set of natural numbers constructed in Volume II, and whose symbols are interpreted by the natural-number addition and natural-number multiplication: $S^{\mathcal{N}}$ is successor, $+^{\mathcal{N}}$ is natural-number addition, $\cdot^{\mathcal{N}}$ is natural-number multiplication, and $0^{\mathcal{N}}, 1^{\mathcal{N}}$ are the natural-number constants 0 and 1.

Remark (Standard quantified statement).

$$|\mathcal{N}| = \mathbb{N}, \qquad S^{\mathcal{N}} : \mathbb{N} \to \mathbb{N}, \qquad +^{\mathcal{N}}, \cdot^{\mathcal{N}} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad 0^{\mathcal{N}}, 1^{\mathcal{N}} \in \mathbb{N}.$$

Remark (Interpretation). Nothing here is new mathematics: the carrier, successor, addition, and multiplication are exactly the objects built in Volume II. The model only packages them as a single object $\mathcal{N}$, so that "$\mathbb{N}$" can be handed to any theorem that quantifies over arithmetic structures. The signature is the type; $\mathcal{N}$ is the instance.

Remark (Satisfaction). $\mathcal{N}$ is a *model* of the natural-number axioms:

$$\mathcal{N} \models \Sigma_{\mathbb{N}}, \qquad \text{equivalently} \qquad \mathrm{PeanoSystem}(\mathbb{N}, S, 0).$$

This is not re-proved here. It is exactly the content of $\mathbb{N}$ Is a Peano System in Volume II, which verifies the Peano axioms against the constructed successor operation. The algebraic arithmetic fragment is witnessed separately by $\mathbb{N}$ Is a Commutative Semiring, which verifies the laws of addition and multiplication on $\mathbb{N}$.

Remark (Expansion to the ordered natural-number model). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{N}_{<} = (\mathbb{N}; S, +, \cdot, 0, 1, <), \qquad \mathrm{Expansion}(\mathcal{N}_{<}, \mathcal{N}),$$

in the sense of Structure Expansion: the carrier is unchanged and exactly one new interpreted symbol is added. Here $<^{\mathcal{N}}$ is the usual order on $\mathbb{N}$. That this expansion has the expected order behavior is witnessed by $\mathbb{N}$ Is Well-Ordered in Volume II.

Remark (Reduct to the successor structure). Forgetting $+, \cdot, 1$ leaves the successor reduct

$$\mathcal{N} \restriction \{S, 0\} = (\mathbb{N}; S, 0), \qquad \mathrm{Reduction}\big(\mathcal{N} \restriction \{S, 0\}, \mathcal{N}\big),$$

in the sense of Structure Reduction: the same carrier with fewer visible symbols. This reduct is the Peano-system view of the natural numbers.

Remark (Reduct to the additive monoid). Forgetting $S, \cdot, 1$ leaves the additive reduct

$$\mathcal{N} \restriction \{+, 0\} = (\mathbb{N}; +, 0), \qquad \mathrm{Reduction}\big(\mathcal{N} \restriction \{+, 0\}, \mathcal{N}\big).$$

This reduct is the additive commutative monoid of natural numbers: it remembers addition and the additive identity, but forgets multiplication and successor as primitive symbols.

Remark (Reduct to the multiplicative monoid). Forgetting $S, +, 0$ leaves the multiplicative reduct

$$\mathcal{N} \restriction \{\cdot, 1\} = (\mathbb{N}; \cdot, 1), \qquad \mathrm{Reduction}\big(\mathcal{N} \restriction \{\cdot, 1\}, \mathcal{N}\big).$$

This reduct is the multiplicative commutative monoid of natural numbers.

Remark (Blueprint position). In the structural blueprint $\mathcal{N}$ is the initial counting structure: it is the first model in the number-system hierarchy and the source of the successor, addition, multiplication, and induction principles used later. It sits at the head of the canonical embedding chain

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The first arrow is the canonical embedding of Volume II, $\mathbb{N}$ Embeds in $\mathbb{Z}$. Passing from $\mathbb{N}$ to $\mathbb{Z}$ supplies additive inverses; passing from $\mathbb{Z}$ to $\mathbb{Q}$ supplies multiplicative inverses for nonzero elements; passing from $\mathbb{Q}$ to $\mathbb{R}$ supplies the analytic completeness needed for real analysis.

Remark (Interpretation). $\mathbb{N}$ is not a group, ring, or field under its usual arithmetic operations: subtraction and division are not closed operations on all of $\mathbb{N}$. Its natural algebraic home is the Peano-system and commutative-semiring layer. This is exactly why the number-system hierarchy proceeds from $\mathbb{N}$ to $\mathbb{Z}$, then to $\mathbb{Q}$, and finally to $\mathbb{R}$.

Remark (Dependencies).

- Arithmetic Signature.
- Peano System.
- The Natural Numbers.
- Addition on $\mathbb{N}$.
- Multiplication on $\mathbb{N}$.

The Whole Numbers as a Structure

Remark (Modeling Goal). We exhibit the whole numbers as a concrete structure with carrier $\mathbb{W} = \{0, 1, 2, \ldots\}$. In the usual convention this is extensionally the same carrier as $\mathbb{N}$ when $\mathbb{N}$ is zero-based. It is kept here as a named presentation because elementary arithmetic often distinguishes "natural numbers" from "whole numbers" terminologically.

Remark (Exposition). The whole-number signature is the zero-based arithmetic fragment

$$\mathcal{L}_{\mathbb{W}} = \{S, +, \cdot, 0, 1\}.$$

It has unary successor S, binary operations $+, \cdot$, and constants $0, 1$. The order symbol $<$ is absent at first and is added by expansion below.

Definition 1.11.3 (The Whole Number Model $\mathcal{W}$)**.** The **whole number model** is the $\mathcal{L}_{\mathbb{W}}$-structure

$$\mathcal{W} = \left(\mathbb{W};\, S^{\mathcal{W}}, +^{\mathcal{W}}, \cdot^{\mathcal{W}}, 0^{\mathcal{W}}, 1^{\mathcal{W}}\right),$$

where

$$\mathbb{W} = \{0, 1, 2, \ldots\},$$

$S^{\mathcal{W}}$ is successor, $+^{\mathcal{W}}$ is whole-number addition, $\cdot^{\mathcal{W}}$ is whole-number multiplication, and $0^{\mathcal{W}}, 1^{\mathcal{W}}$ are the distinguished whole-number constants.

Remark (Standard quantified statement).

$$|\mathcal{W}| = \mathbb{W}, \qquad S^{\mathcal{W}} : \mathbb{W} \to \mathbb{W}, \qquad +^{\mathcal{W}}, \cdot^{\mathcal{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}, \qquad 0^{\mathcal{W}}, 1^{\mathcal{W}} \in \mathbb{W}.$$

Remark (Interpretation). If the repository defines $\mathbb{N}$ as $\{0, 1, 2, \ldots\}$, then $\mathcal{W}$ is not a new mathematical structure; it is the same zero-based counting structure under the name "whole numbers." If the repository defines $\mathbb{N}$ as $\{1, 2, 3, \ldots\}$, then $\mathbb{W}$ is the zero-adjoined successor structure.

Remark (Satisfaction). The whole-number model satisfies the same zero-based Peano and arithmetic laws as the natural-number model whenever the repository convention identifies

$$\mathbb{W} = \mathbb{N}.$$

Equivalently,

$$\mathcal{W} \models \Sigma_{\mathbb{W}}, \qquad \mathrm{PeanoSystem}(\mathbb{W}, S, 0).$$

This satisfaction claim should be wired to the repository theorem that verifies the Peano-system or commutative-semiring laws for the zero-based counting structure.

Remark (Expansion to the ordered whole-number model). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{W}_< = (\mathbb{W}; S, +, \cdot, 0, 1, <), \qquad \mathrm{Expansion}(\mathcal{W}_<, \mathcal{W}),$$

where $<^{\mathcal{W}}$ is the usual order on $\{0, 1, 2, \ldots\}$.

Remark (Reduct to the successor structure). Forgetting $+, \cdot, 1$ leaves the successor reduct

$$\mathcal{W}\restriction\{S, 0\} = (\mathbb{W}; S, 0), \qquad \mathrm{Reduction}\big(\mathcal{W}\restriction\{S, 0\},\, \mathcal{W}\big).$$

This is the zero-based counting structure.

Remark (Reduct to the additive monoid). Forgetting $S, \cdot, 1$ leaves

$$\mathcal{W}\restriction\{+, 0\} = (\mathbb{W}; +, 0),$$

the additive commutative monoid of whole numbers.

Remark (Blueprint position). The whole numbers occupy a terminological rather than algebraically new position. They emphasize that zero is included:

$$\mathbb{W} = \{0, 1, 2, \ldots\}.$$

Structurally, they sit at the zero-based counting layer and embed into the integers:

$$\mathbb{W} \hookrightarrow \mathbb{Z}.$$

Passing from $\mathbb{W}$ to $\mathbb{Z}$ supplies additive inverses.

Remark (Interpretation). The whole numbers are closed under addition and multiplication, contain 0 and 1, and support successor. They are not a group, ring, or field under the usual arithmetic operations because additive inverses and multiplicative inverses are not available internally. Their natural algebraic home is the same commutative-semiring / zero-based Peano layer as the natural-number model.

The Integers as a Structure

Remark (Modeling Goal). We exhibit $\mathbb{Z}$ as a concrete *model*: fix the ring signature, take the carrier and operations constructed in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{Z}$ satisfies the ring axioms. Order is then added by expansion, and various algebraic fragments are recovered by reduction.

Remark (Exposition). A model is a signature together with an interpretation on a carrier. The signature here is the ring fragment of the arithmetic signature,

$$\mathcal{L}_{\text{ring}} = \{+, \cdot, -, 0, 1\},$$

with binary function symbols $+, \cdot$, unary function symbol $-$, and constant symbols $0, 1$.

The order symbol $<$ is deliberately absent. It enters only through expansion.

Definition 1.11.4 (The Integer Model $\mathcal{Z}$)**.** The **integer model** is the $\mathcal{L}_{\text{ring}}$-structure

$$\mathcal{Z} = \big(\mathbb{Z}; +^{\mathcal{Z}}, \cdot^{\mathcal{Z}}, -^{\mathcal{Z}}, 0^{\mathcal{Z}}, 1^{\mathcal{Z}}\big),$$

whose carrier $|\mathcal{Z}| = \mathbb{Z}$ is the set of integers constructed in Volume II, and whose symbols are interpreted by the integer addition and integer multiplication.

Thus

$$+^{\mathcal{Z}}$$

is integer addition,

$$\cdot^{\mathcal{Z}}$$

is integer multiplication,

$$-^{\mathcal{Z}}$$

is integer negation, and

$$0^{\mathcal{Z}}, \quad 1^{\mathcal{Z}}$$

are the distinguished integer constants.

Remark (Standard quantified statement)*.*

$$|\mathcal{Z}| = \mathbb{Z},$$

$$+^{\mathcal{Z}}, \cdot^{\mathcal{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},$$

$$-^{\mathcal{Z}} : \mathbb{Z} \to \mathbb{Z},$$

$$0^{\mathcal{Z}}, 1^{\mathcal{Z}} \in \mathbb{Z}.$$

Remark (Interpretation)*.* Nothing new is constructed here.

The carrier and operations were already built in Volume II.

The purpose of the model

$$\mathcal{Z}$$

is to package the integer system as a single mathematical object so that theorems may quantify over rings and then be applied directly to the integers.

The signature is the type.

The structure

$$\mathcal{Z}$$

is the instance.

Remark (Satisfaction). The structure

$$\mathcal{Z}$$

is a model of the ring axioms:

$$\mathcal{Z} \models \Sigma_{\text{ring}},$$

equivalently,

$$\text{Ring}(\mathbb{Z}, +, \cdot, 0, 1).$$

This is not re-proved here.
It is exactly the content of
$\mathbb{Z}$ Is a Ring
in Volume II, which verifies each ring axiom against the integer operations.
Likewise,

$$\mathcal{Z} \models \Sigma_{\text{id}},$$

because
$\mathbb{Z}$ Is an Integral Domain
shows that multiplication on the integers has no zero divisors.

Remark (Expansion to the Ordered Ring). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{Z}_< = (\mathbb{Z}; +, \cdot, -, 0, 1, <),$$

$$\text{Expansion}(\mathcal{Z}_<, \mathcal{Z}),$$

in the sense of Structure Expansion.
The carrier is unchanged and exactly one interpreted symbol is added.
The relation

$$<^{\mathcal{Z}}$$

is the usual integer order determined by the positive cone.
That this expansion satisfies the ordered-ring axioms is witnessed by
$\mathbb{Z}$ Is an Ordered Ring
in Volume II.

Remark (Reduct to the Additive Group). Forgetting

$$\cdot, 1$$

leaves the reduct

$$\mathcal{Z} \restriction \{+, -, 0\} = (\mathbb{Z}; +, -, 0),$$

$$\text{Reduction}\big(\mathcal{Z} \restriction \{+, -, 0\}, \mathcal{Z}\big).$$

This reduct is the additive abelian group of integers.
It remembers addition and additive inverses while ignoring multiplication.

Remark (Reduct to the Multiplicative Monoid). Forgetting

$$+, -, 0$$

leaves

$$\mathcal{Z} \restriction \{\cdot, 1\} = (\mathbb{Z}; \cdot, 1),$$

the multiplicative commutative monoid of integers.

Remark (Blueprint Position). The integer model occupies the next stage after the natural numbers in the number-system hierarchy.

It sits on the embedding chain

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The first embedding is the canonical map

$$n \longmapsto n,$$

witnessed by
$\mathbb{N}$ Embeds in $\mathbb{Z}$.
The second embedding is witnessed by
$\mathbb{Z}$ Embeds in $\mathbb{Q}$.

Remark (Interpretation). The passage

$$\mathbb{N} \longrightarrow \mathbb{Z}$$

supplies exactly one new algebraic capability:

$$\forall a \in \mathbb{Z}, \quad \exists(-a) \in \mathbb{Z}.$$

Every integer has an additive inverse.
This is the structural feature absent from the natural numbers.
Consequently,

$$(\mathbb{N}, +)$$

is only a commutative monoid, whereas

$$(\mathbb{Z}, +)$$

is an abelian group.
The passage

$$\mathbb{Z} \longrightarrow \mathbb{Q}$$

will later supply multiplicative inverses for nonzero elements and thereby upgrade the integer ring into a field.

Remark (Structural Significance). The integer model is the first standard number system that satisfies the full ring signature.

The natural numbers support addition and multiplication, but not additive inverses.
The integers close precisely that gap.
For this reason $\mathbb{Z}$ occupies the ring layer of the structural hierarchy, sitting between the semiring structure of $\mathbb{N}$ and the field structure of $\mathbb{Q}$.

Remark (Dependencies).

- Arithmetic Signature.
- Ring.
- The Integers.
- Integer Addition.
- Integer Multiplication.

The Rational Numbers as a Structure

Remark (Modeling Goal). We exhibit $\mathbb{Q}$ as a concrete *model*: fix the field signature, take the carrier and operations built in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{Q}$ obeys the field axioms. Order is then added by *expansion*, and the additive group is recovered by *reduction*.

Remark (Exposition). A model is a signature together with an interpretation on a carrier. The signature here is the field fragment of the arithmetic signature,

$$\mathcal{L}_{\text{field}} = \{+, \cdot, -, {}^{-1}, 0, 1\},$$

with binary function symbols $+, \cdot$, a unary $-$, a unary ${}^{-1}$ defined on nonzero elements, and constant symbols $0, 1$. The order symbol $<$ is deliberately absent; it enters by expansion below.

Definition 1.11.5 (The Rational Field Model $\mathcal{Q}$)**.** The **rational field model** is the $\mathcal{L}_{\text{field}}$-structure

$$\mathcal{Q} = \big(\mathbb{Q};\ +^{\mathcal{Q}},\ \cdot^{\mathcal{Q}},\ -^{\mathcal{Q}},\ {}^{-1\,\mathcal{Q}},\ 0^{\mathcal{Q}},\ 1^{\mathcal{Q}}\big),$$

whose carrier $|\mathcal{Q}| = \mathbb{Q}$ is the set of rational numbers constructed in Volume II, and whose symbols are interpreted by the rational addition and rational multiplication: $+^{\mathcal{Q}}$ and $\cdot^{\mathcal{Q}}$ are rational addition and multiplication, $-^{\mathcal{Q}}$ is rational negation, ${}^{-1\,\mathcal{Q}}$ is the inverse of a nonzero rational, and $0^{\mathcal{Q}}, 1^{\mathcal{Q}}$ are the rational constants 0 and 1.

Remark (Standard quantified statement).

$$|\mathcal{Q}| = \mathbb{Q}, \qquad +^{\mathcal{Q}}, \cdot^{\mathcal{Q}} : \mathbb{Q}\times\mathbb{Q} \to \mathbb{Q}, \qquad -^{\mathcal{Q}} : \mathbb{Q} \to \mathbb{Q}, \qquad 0^{\mathcal{Q}}, 1^{\mathcal{Q}} \in \mathbb{Q}.$$

Remark (Interpretation). Nothing here is new mathematics: the carrier and operations are exactly the ones built in Volume II. The model only packages them as a single object $\mathcal{Q}$, so that "$\mathbb{Q}$" can be handed to any theorem that quantifies over fields, and so the satisfaction statement below has a concrete structure to range over. The signature is the type; $\mathcal{Q}$ is the instance.

Remark (Satisfaction). $\mathcal{Q}$ is a *model* of the field axioms:

$$\mathcal{Q} \models \Sigma_{\text{field}}, \qquad \text{equivalently} \qquad \mathrm{Field}(\mathbb{Q}, +, \cdot, 0, 1).$$

This is not re-proved here. It is exactly the content of $\mathbb{Q}$ Is a Field in Volume II, which verifies every field axiom against the rational operations; that theorem is therefore the witness that the interpretation $\mathcal{Q}$ satisfies Σ_{field}. The integral-domain fragment is witnessed separately by $\mathbb{Q}$ Is an Integral Domain.

Remark (Expansion to the ordered field). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{Q}_< = (\mathbb{Q}; +, \cdot, -, {}^{-1}, 0, 1, <), \qquad \mathrm{Expansion}(\mathcal{Q}_<, \mathcal{Q}),$$

in the sense of Structure Expansion: the carrier is unchanged and exactly one new interpreted symbol is added, where $<^{\mathcal{Q}}$ is the rational order determined by the positive cone. That $\mathcal{Q}_<$ satisfies the ordered-field axioms is witnessed by $\mathbb{Q}$ Is an Ordered Field in Volume II.

Remark (Reduct to the additive group). Forgetting $\cdot, {}^{-1}, 1$ leaves the reduct

$$\mathcal{Q}{\restriction}\{+, -, 0\} = (\mathbb{Q}; +, -, 0), \qquad \mathrm{Reduction}\big(\mathcal{Q}{\restriction}\{+, -, 0\},\ \mathcal{Q}\big),$$

in the sense of Structure Reduction: the same carrier with fewer visible symbols. This reduct is the additive abelian group of $\mathbb{Q}$ — what remains when only additive structure is in view.

Remark (Blueprint position). In the structural blueprint $\mathcal{Q}$ is the *field of fractions* of the integer ring: the smallest field into which the integer model embeds. This places it on the chain

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R},$$

whose arrows are the canonical embeddings of Volume II ($\mathbb{Z}$ Embeds in $\mathbb{Q}$, $\mathbb{Q}$ Embeds in $\mathbb{R}$). Model-theoretically the first embedding presents the integer model as a substructure of the $\{+, \cdot, 0, 1\}$-reduct of $\mathcal{Q}$; passing to the full field $\mathcal{Q}$ supplies the multiplicative inverses the integer model lacks.

Remark (Interpretation). $\mathbb{Q}$ is the first number system whose model satisfies the *full* field signature: $\mathbb{Z}$ is only a ring (no inverses), and $\mathbb{Q}$ closes exactly that gap. Adding $<$ by expansion gives the ordered field; adding analytic completeness — which is *not* a first-order property of the field signature — is deferred to the real model in the analysis layer.

Remark (Dependencies).

- Arithmetic Signature.
- Field (Volume II).
- The Rational Numbers.
- Rational Addition.
- Rational Multiplication.

The Real Numbers as a Structure

Remark (Modeling Goal). We exhibit $\mathbb{R}$ as a concrete model of the ordered-field signature. The field structure is first-order and is captured by satisfaction of the ordered-field axioms. The completeness of $\mathbb{R}$ is then discussed separately, because it does not belong to the ordinary first-order field signature.

Remark (Exposition)*.* A model consists of a signature together with an interpretation on a carrier.

The signature considered here is the ordered-field signature

$$\mathcal{L}_{\mathrm{of}} = \{+, \cdot, -, {}^{-1}, 0, 1, <\}.$$

It contains the field operations together with a distinguished order relation.

Definition 1.11.6 (The Real Number Model $\mathcal{R}$)**.** The **real number model** is the $\mathcal{L}_{\mathrm{of}}$-structure

$$\mathcal{R} = \left(\mathbb{R}; +^{\mathcal{R}}, \cdot^{\mathcal{R}}, -^{\mathcal{R}}, {}^{-1\,\mathcal{R}}, 0^{\mathcal{R}}, 1^{\mathcal{R}}, <^{\mathcal{R}}\right),$$

whose carrier

$$|\mathcal{R}| = \mathbb{R}$$

is the set of real numbers constructed in Volume II, and whose symbols are interpreted by the real-number operations and order.

Remark (Standard quantified statement)*.*

$$|\mathcal{R}| = \mathbb{R},$$

$$+^{\mathcal{R}}, \cdot^{\mathcal{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R},$$

$$-^{\mathcal{R}} : \mathbb{R} \to \mathbb{R},$$

$${}^{-1\,\mathcal{R}} : \mathbb{R} \setminus \{0\} \to \mathbb{R},$$

$$0^{\mathcal{R}}, 1^{\mathcal{R}} \in \mathbb{R},$$

$$<^{\mathcal{R}} \subseteq \mathbb{R} \times \mathbb{R}.$$

Remark (Interpretation)*.* The structure

$$\mathcal{R}$$

packages the entire real-number system into a single mathematical object.

The carrier, operations, inverses, and order already exist from the Volume II construction. The model simply presents them as a unified structure that may serve as an argument to general theorems about ordered fields.

Remark (Satisfaction)*.* The structure

$$\mathcal{R}$$

is a model of the ordered-field axioms:

$$\mathcal{R} \models \Sigma_{\mathrm{of}},$$

equivalently,

$$\mathrm{OrderedField}(\mathbb{R}, +, \cdot, <, 0, 1).$$

This is witnessed by
$\mathbb{R}$ Is an Ordered Field
in Volume II.
In particular,

$$\mathcal{R} \models \Sigma_{\text{field}}$$

because
$\mathbb{R}$ Is a Field
verifies the field axioms.

Remark (Completeness Is Separate)*.* The ordered-field structure does *not* yet distinguish $\mathbb{R}$ from every other ordered field.

The defining analytic property of the real numbers is completeness, expressed in Volume II by

$$LeastUpperBoundProperty(\mathbb{R}).$$

Equivalently, every nonempty subset of $\mathbb{R}$ that is bounded above possesses a least upper bound.

This property is witnessed by
Completeness of the Real Numbers.

Remark (Important Structural Distinction)*.* The statement

$$\mathcal{R} \models \Sigma_{\text{of}}$$

belongs to the first-order ordered-field structure.
The completeness property does not belong to the ordinary ordered-field signature.
Consequently the real-number model should be viewed as

$$\text{ordered field} + \text{completeness}.$$

These are conceptually distinct layers.

Remark (Reduction to the Field Structure)*.* Forgetting the order relation leaves the reduct

$$\mathcal{R} \restriction \{+, \cdot, -, {}^{-1}, 0, 1\},$$

which is simply the field structure of the real numbers.

$$\mathrm{Reduction}\Big(\mathcal{R} \restriction \{+, \cdot, -, {}^{-1}, 0, 1\}, \mathcal{R}\Big).$$

Remark (Reduction to the Additive Group)*.* Forgetting multiplication and order leaves

$$(\mathbb{R}; +, -, 0),$$

the additive abelian group of real numbers.

Remark (Blueprint Position)*.* The real-number model occupies the final stage of the standard ordered number-system hierarchy:

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The final embedding is witnessed by
$\mathbb{Q}$ Embeds in $\mathbb{R}$.
The rationals appear inside $\mathbb{R}$ as a subfield.

Remark (Interpretation)*.* The passage

$$\mathbb{Q} \longrightarrow \mathbb{R}$$

does not add new field operations.
Addition, subtraction, multiplication, and division already exist in $\mathbb{Q}$.
Instead it adds completeness.
This closes the gaps left by the rational numbers and supplies limits, suprema, infima, and the analytic foundation required for calculus and real analysis.

Remark (Structural Significance)*.* The real-number model is simultaneously

- a field,
- an ordered field,
- a complete ordered field.

The first two properties belong to its algebraic structure.
The third is the property that makes real analysis possible.
For this reason $\mathbb{R}$ occupies the analytic layer of the structural hierarchy and serves as the foundational model for limits, continuity, differentiation, and integration.

Remark (Dependencies)*.*

- Field.
- Positive Cone.
- The Real Numbers.
- $\mathbb{R}$ Is a Field.
- $\mathbb{R}$ Is an Ordered Field.
- $\mathbb{R}$ Is a Complete Ordered Field.

The Complex Numbers as a Structure

Remark (Modeling Goal)*.* We exhibit $\mathbb{C}$ as a concrete field model. The real numbers embed into $\mathbb{C}$, and the resulting structure is the first standard number system that is algebraically closed. Unlike $\mathbb{R}$, however, $\mathbb{C}$ admits no compatible ordered-field expansion.

Remark (Exposition). A model consists of a signature together with an interpretation on a carrier.

The signature considered here is the field signature

$$\mathcal{L}_{\text{field}} = \{+, \cdot, -, {}^{-1}, 0, 1\}.$$

It contains the field operations and distinguished constants but no order relation.

Definition 1.11.7 (The Complex Number Model $\mathcal{C}$)**.** The **complex number model** is the $\mathcal{L}_{\text{field}}$-structure

$$\mathcal{C} = \left(\mathbb{C}; +^{\mathcal{C}}, \cdot^{\mathcal{C}}, -^{\mathcal{C}}, {}^{-1\mathcal{C}}, 0^{\mathcal{C}}, 1^{\mathcal{C}}\right),$$

whose carrier

$$|\mathcal{C}| = \mathbb{C}$$

is the set of complex numbers constructed in Volume II, and whose symbols are interpreted by the usual complex-number operations.

Remark (Standard quantified statement).

$$|\mathcal{C}| = \mathbb{C},$$

$$+^{\mathcal{C}}, \cdot^{\mathcal{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C},$$

$$-^{\mathcal{C}} : \mathbb{C} \to \mathbb{C},$$

$${}^{-1\mathcal{C}} : \mathbb{C} \setminus \{0\} \to \mathbb{C},$$

$$0^{\mathcal{C}}, 1^{\mathcal{C}} \in \mathbb{C}.$$

Remark (Interpretation). The structure

$$\mathcal{C}$$

packages the complex-number system into a single mathematical object.

The carrier and operations are those already constructed in Volume II. The model merely records how the field signature is interpreted on that carrier.

Remark (Satisfaction). The structure

$$\mathcal{C}$$

is a model of the field axioms:

$$\mathcal{C} \models \Sigma_{\text{field}},$$

equivalently,

$$\text{Field}(\mathbb{C}, +, \cdot, 0, 1).$$

This is witnessed by
$\mathbb{C}$ Is a Field
in Volume II.

Remark (Algebraic Closure). The most important structural property of the complex numbers is not merely that they form a field.

The defining feature of $\mathbb{C}$ is that it is algebraically closed.

Equivalently,

$$\forall p(x) \in \mathbb{C}[x], \qquad \deg p \geq 1 \Longrightarrow \exists z \in \mathbb{C} \quad p(z) = 0.$$

This property is witnessed by

Fundamental Theorem of Algebra.

Remark (Interpretation). Every nonconstant polynomial with complex coefficients has a complex root.

Consequently every polynomial factors completely into linear factors over $\mathbb{C}$.

This is the structural reason the complex numbers occupy the final stage of the standard algebraic number-system hierarchy.

Remark (Real Numbers as a Substructure). The real numbers embed into the complex numbers through the canonical map

$$r \longmapsto r + 0i.$$

This embedding is witnessed by

$\mathbb{R}$ Embeds in $\mathbb{C}$.

Under this embedding the real numbers appear as a subfield of the complex field.

Remark (Reduction to the Additive Group). Forgetting multiplication leaves

$$(\mathbb{C}; +, -, 0),$$

the additive abelian group of complex numbers.

Remark (Reduction to the Multiplicative Group). Restricting to nonzero elements and forgetting addition leaves

$$(\mathbb{C}^{\times}; \cdot, 1),$$

the multiplicative abelian group of nonzero complex numbers.

Remark (No Ordered-Field Expansion). Unlike $\mathbb{Q}$ and $\mathbb{R}$, the complex field cannot be expanded into an ordered field compatible with its field operations.

Indeed,

$$i^2 = -1.$$

In every ordered field, every square is nonnegative. Hence

$$i^2 \geq 0.$$

But this would imply

$$-1 \geq 0,$$

which contradicts the ordered-field axioms.

Therefore no order relation can be added to the complex field so as to make it an ordered field.

Remark (Blueprint Position)*.* The complex-number model occupies the final stage of the standard number hierarchy:

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C}.$$

The passage

$$\mathbb{R} \longrightarrow \mathbb{C}$$

supplies solutions to polynomial equations that have no real solutions.
For example,

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$ but has the solutions

$$x = \pm i$$

in $\mathbb{C}$.

Remark (Structural Significance)*.* The complex-number model is

- a field,
- an algebraically closed field,
- not an ordered field.

The real numbers complete the ordered-field hierarchy.
The complex numbers complete the algebraic hierarchy.
For this reason $\mathbb{C}$ serves as the canonical ambient field for algebra, complex analysis, algebraic geometry, and much of modern mathematics.

Remark (Dependencies)*.*

- Field.
- The Complex Numbers.
- Addition on $\mathbb{C}$.
- Multiplication on $\mathbb{C}$.
- $\mathbb{C}$ Is a Field.
- TODO: formalize the Fundamental Theorem of Algebra.

1.12 The Number Systems as $\mathcal{L}$-Structures

The Arithmetic Signature

The number systems $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ all live in the same signature — they differ in which axiom *sentences* they satisfy, not in which symbols they carry.

The Arithmetic Signature $\mathcal{L}_{\text{arith}}$

$$\mathcal{L}_{\text{arith}} = \{ \underbrace{+, \cdot,}_{\text{arity 2}} \ \underbrace{-, (\cdot)^{-1},}_{\text{arity 1}} \ \underbrace{0, 1}_{\text{constants}}, \ \underbrace{\leq}_{\text{relation, arity 2}} \}.$$

Symbol	Arity	Intended interpretation
$+$	2 (function)	addition
$\cdot$	2 (function)	multiplication
$-$	1 (function)	additive inverse / negation
$(\cdot)^{-1}$	1 (function)	multiplicative inverse
0	0 (constant)	additive identity
1	0 (constant)	multiplicative identity
$\leq$	2 (relation)	order

Remark 1.12.1. Every number system is an $\mathcal{L}_{\text{arith}}$-structure: it carries all these symbols. What distinguishes $\mathbb{N}$ from $\mathbb{Z}$ from $\mathbb{R}$ is not the signature but the *theory* — the set of first-order sentences in $\mathcal{L}_{\text{arith}}$ that hold in each model.

In particular, $(\cdot)^{-1}$ is a symbol in the signature of $\mathbb{Z}$, but the sentence

$$\forall x \ (x \neq 0 \Rightarrow x \cdot x^{-1} = 1)$$

is *false* in $\mathbb{Z}$. The symbol is interpreted as a total function on $\mathbb{Z}$ (defined arbitrarily on 0 and on nonzero elements where the true inverse does not exist), but the field axiom it satisfies in $\mathbb{Q}$ and $\mathbb{R}$ fails. This is the model-theoretic way of saying: $\mathbb{Z}$ is not a field.

Each Number System as a Model

Number Systems as Models of $\mathcal{L}_{\text{arith}}$

Structure	Carrier	Theory (axiom sentences satisfied)
$\mathcal{N} = (\mathbb{N}, +, \cdot, -, (\cdot)^{-1}, 0, 1, \leq)$	$\mathbb{N}$	Semiring axioms + well-ordering. $-$ and $(\cdot)^{-1}$ defined but inverse
$\mathcal{Z} = (\mathbb{Z}, +, \cdot, -, (\cdot)^{-1}, 0, 1, \leq)$	$\mathbb{Z}$	Commutative ordered ring axioms. Additive inverse axiom holds;
$\mathcal{Q} = (\mathbb{Q}, +, \cdot, -, (\cdot)^{-1}, 0, 1, \leq)$	$\mathbb{Q}$	Ordered field axioms. Both inverse axioms hold; least upper bour
$\mathcal{R} = (\mathbb{R}, +, \cdot, -, (\cdot)^{-1}, 0, 1, \leq)$	$\mathbb{R}$	Complete ordered field axioms. All axioms hold; $\mathbb{R}$ is the unique

Remark 1.12.2 (The abstract algebra framing of the extension hierarchy). From the blueprint hierarchy in this chapter:

$$\underbrace{\mathbb{N}}_{\text{semiring}} \subset \underbrace{\mathbb{Z}}_{\text{comm. ring}} \subset \underbrace{\mathbb{Q}}_{\text{ordered field}} \subset \underbrace{\mathbb{R}}_{\text{complete ordered field}}.$$

Each inclusion is a *substructure embedding*: the smaller system embeds into the larger as an $\mathcal{L}_{\text{arith}}$-substructure (operations and constants are preserved, but more sentences become true in the larger model).

More precisely: $\mathbb{N} \hookrightarrow \mathbb{Z}$ adds the solution $-n$ to every equation $n + x = 0$. $\mathbb{Z} \hookrightarrow \mathbb{Q}$ adds the solution p/q to every equation $qx = p$ with $q \neq 0$. $\mathbb{Q} \hookrightarrow \mathbb{R}$ adds a limit point to every bounded increasing sequence — it is a *completion*, not a purely algebraic extension.

The Blueprint Position of Each Number System

Number Systems in the Blueprint Hierarchy

System	Blueprint level	What it adds
$\mathbb{N}$	Commutative semiring	Two operations, distributivity, no additive inverses.
$\mathbb{Z}$	Commutative ordered ring	Additive inverses (Blueprint 8 = Ring, commutative).
$\mathbb{Q}$	Ordered field	Multiplicative inverses (Blueprint 9 = Field).
$\mathbb{R}$	Complete ordered field	Least upper bound property (analytic, not purely algebraic).

Remark 1.12.3 (Where completeness lives). The least upper bound property is *not* expressible as a single first-order sentence in $\mathcal{L}_{\text{arith}}$ — it is a second-order axiom schema (quantifying over subsets). This is why $\mathbb{R}$ cannot be characterised up to isomorphism by any first-order theory: by the compactness theorem, any first-order theory satisfied by $\mathbb{R}$ also has non-standard models (the hyperreals). The categorical characterisation — $\mathbb{R}$ is the unique complete ordered field up to isomorphism — requires the full second-order least upper bound axiom. This is the boundary between algebra and analysis.

Remark 1.12.4 (Derived properties are sentences, not axioms). Properties like $a \cdot 0 = 0$, $(-a)b = -(ab)$, and $ab = 0 \Rightarrow a = 0 \vee b = 0$ (no zero divisors in $\mathbb{Z}$) are *theorems*: first-order sentences derivable from the ring axioms. They hold in every model of those axioms, not just in $\mathbb{Z}$. From the model-theoretic perspective, the ring axioms are a theory T, and these properties are elements of the *deductive closure* $\mathrm{Th}(T)$ — they are true in every model of T. See Volume II §**??** for the concrete derivations.

1.13 Number Systems as Models

Each Number System as a Model

Remark (Exposition). Each number system is a model once a signature is fixed. The carrier supplies the elements, and the operations, relations, and constants interpret the symbols of the chosen arithmetic language.

Structure Expansion

Definition 1.13.1 (Structure Expansion)**.** A **structure expansion** is obtained by adding new interpreted symbols to an existing structure while keeping the same carrier set.

Remark (Standard quantified statement).

$$\text{Expansion}(\mathcal{A}', \mathcal{A}) \iff |\mathcal{A}'| = |\mathcal{A}| \text{ and } \mathcal{A}' \text{ interprets all symbols of } \mathcal{A} \text{ plus additional symbols.}$$

Remark (Definition predicate reading). $\mathcal{A}'$ expands $\mathcal{A}$ exactly when it keeps the same carrier and adds visible structure.

Remark (Negated quantified statement).

$$\neg\,\text{Expansion}(\mathcal{A}', \mathcal{A})$$

when the carrier changes or the original interpretations are not preserved.

Remark (Interpretation). Adding an order relation to a field signature expands the visible structure: the carrier remains the same, but more predicates or operations are available.

Remark (Dependencies).

- Carrier Set
- First-Order Signature

Structure Reduction

Definition 1.13.2 (Structure Reduction)**.** A **structure reduction** is obtained by forgetting some interpreted symbols of a structure while keeping the same carrier set.

Remark (Standard quantified statement).

$$\text{Reduction}(\mathcal{A}_0, \mathcal{A}) \iff |\mathcal{A}_0| = |\mathcal{A}| \text{ and } \mathcal{A}_0 \text{ retains only selected symbols of } \mathcal{A}.$$

Remark (Definition predicate reading). $\mathcal{A}_0$ reduces $\mathcal{A}$ exactly when it keeps the carrier and forgets some visible structure.

Remark (Negated quantified statement).

$$\neg\,\text{Reduction}(\mathcal{A}_0, \mathcal{A})$$

when the carrier changes or the retained interpretations do not agree.

Remark (Interpretation). Forgetting multiplication in a ring leaves its additive group visible. Forgetting order in an ordered field leaves the underlying field structure.

Remark (Dependencies).

- Carrier Set
- First-Order Signature

Blueprint Position of Each Number System

Remark (Exposition). The standard number systems occupy different positions in the algebraic blueprint: $\mathbb{Z}$ is a ring, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields, and $\mathbb{C}$ is a field without the usual compatible total order.

Remark (Regeneration Note). The prior working notes contain broad hierarchy prose, but a final table should be reconciled with the repository's canonical number-system conventions.

Chapter 2

Abstract Algebra

abstract-algebra

Algebra → **Abstract Algebra** → Formalizing Algebraic Structures

Chapter 3

Algebraic Geometry

algebraic-geometry

Algebra $\rightarrow$ **Algebraic Geometry** $\rightarrow$ Formalizing Algebraic Structures

3.1 Algebraic Geometry

3.2 Preliminary Definitions

Definition 3.2.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Remark (Standard quantified statement). A ring is a set equipped with addition and multiplication satisfying the additive-group, multiplicative-associativity, multiplicative-identity, and distributive laws.

Remark (Interpretation). A ring generalizes a field by dropping the requirement that nonzero elements have multiplicative inverses, and by not requiring multiplication to be commutative. When multiplication is commutative, we call R a *commutative ring*.

Definition 3.2.2 (Commutative Ring)**.** A ring R is *commutative* if
$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Remark (Standard quantified statement)*.* A ring R is commutative when multiplication in R is commutative.

Remark (Structural Summary)*.* The hierarchy of algebraic structures encountered so far is:

Structure	Additive Group	Multiplicative Structure
Ring	Abelian	Associative, with identity
Commutative Ring	Abelian	Associative, commutative, with identity
Field	Abelian	Abelian on nonzero elements

Every field is a commutative ring, but not every commutative ring is a field.

Polynomial Rings

Definition 3.2.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression

$$f = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate.*

Remark (Standard quantified statement)*.* A polynomial over R is a finite formal sum $\sum_i a_i x^i$ with coefficients $a_i \in R$.

Remark (Interpretation)*.* The indeterminate x is not a variable ranging over values in R. It is a formal placeholder that encodes the coefficient sequence. Two polynomials are equal if and only if all their coefficients are equal.

Definition 3.2.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is

$$\deg(f) := n.$$

The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic.*

Remark (Standard quantified statement)*.* For a nonzero polynomial $f = a_n x^n + \cdots + a_0$ with leading coefficient $a_n \neq 0$, the degree of f is n.

Remark (Interpretation)*.* The zero polynomial $f = 0$ has no leading coefficient. Its degree is left undefined, or assigned $-\infty$ by convention to preserve the identity $\deg(fg) = \deg(f) + \deg(g)$.

Definition 3.2.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:

$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$

$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Remark (Standard quantified statement). The set $R[x]$ is the collection of finite formal sums in x with coefficients in R, equipped with coefficientwise addition and Cauchy-product multiplication.

Example (Polynomial arithmetic over the integers). Let $R = \mathbb{Z}$ and consider

$$f = 2x^2 + 3x + 1, \qquad g = x + 4 \quad \in \mathbb{Z}[x].$$

Then:

$$f + g = 2x^2 + 4x + 5,$$
$$f \cdot g = 2x^3 + 11x^2 + 13x + 4.$$

Remark (Structural Summary). With these operations, $R[x]$ is itself a commutative ring. The original ring R embeds into $R[x]$ as the constant polynomials.

Property	Holds in $R[x]$?
Commutative ring	Always
Field	Only in degenerate cases
R embeds in $R[x]$	Always

Polynomial Rings in Several Variables

Definition 3.2.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \dots, x_n],$$

is defined inductively by

$$R[x_1, \dots, x_n] := R[x_1, \dots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_{\alpha} a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Remark (Standard quantified statement). The ring $R[x_1, \dots, x_n]$ consists of finite formal sums $\sum_\alpha a_\alpha x^\alpha$ indexed by multi-indices.

Definition 3.2.7 (Monomial)**.** A *monomial* in $R[x_1, \dots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Remark (Standard quantified statement). A monomial in $R[x_1, \dots, x_n]$ is a single multi-index power x^α.

Definition 3.2.8 (Total Degree)**.** The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \dots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Remark (Standard quantified statement). The total degree of x^α is $|\alpha| = \alpha_1 + \cdots + \alpha_n$.

Example (Example). In $\mathbb{R}[x, y, z]$, the polynomial

$$f = 3x^2y + xy^2z - 5z^3$$

has three terms with total degrees 3, 4, and 3 respectively. Hence $\deg(f) = 4$.

Remark (Relevance to Algebraic Geometry). Polynomial rings in several variables are the foundational algebraic object of algebraic geometry. The geometric objects studied in subsequent sections — affine varieties, ideals, coordinate rings — are all defined in terms of $k[x_1, \dots, x_n]$ for a field k. The interplay between the algebra of $k[x_1, \dots, x_n]$ and the geometry of its zero sets is the central theme of Clader–Ross.

Ideals

Definition 3.2.9 (Ideal)**.** Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Remark (Standard quantified statement). An ideal of R is an additive subgroup of R closed under multiplication by arbitrary elements of R.

Definition 3.2.10 (Generated Ideal)**.** Let R be a commutative ring and let $f_1, \dots, f_m \in R$. The *ideal generated by* $f_1, \dots, f_m$ is

$$\langle f_1, \dots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Remark (Standard quantified statement). The ideal generated by $f_1, \dots, f_m$ is the set of all finite R-linear combinations of those generators.

Definition 3.2.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \dots, f_m \in R$ such that $I = \langle f_1, \dots, f_m \rangle$.

Remark (Standard quantified statement). An ideal I is finitely generated when $I = \langle f_1, \dots, f_m \rangle$ for some finite list $f_1, \dots, f_m$ of elements of R.

Definition 3.2.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Remark (Standard quantified statement). A commutative ring R is Noetherian when every ideal $I \subseteq R$ is finitely generated.

Theorem 3.2.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \dots, x_n]$ is Noetherian for any field k. *Go to proof.*

Remark (Standard quantified statement). If R is Noetherian, then $R[x]$ is Noetherian.

Remark (Interpretation). The Hilbert Basis Theorem guarantees that every ideal in $k[x_1, \dots, x_n]$ is finitely generated. This is a foundational finiteness result: it means every algebraic variety can be cut out by finitely many polynomial equations.

Quotient Rings

Definition 3.2.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Remark (Standard quantified statement)*.* For an ideal I of R, the quotient R/I is the set of additive cosets of I with operations induced from R.

Remark (Interpretation)*.* The quotient ring R/I is a commutative ring. Elements of R/I are equivalence classes under the relation $a \sim b \iff a - b \in I$.

Example (A complex quotient)*.* In $\mathbb{R}[x]$, consider the ideal $I = \langle x^2 + 1 \rangle$. The quotient ring

$$\mathbb{R}[x]/\langle x^2 + 1 \rangle \;\cong\; \mathbb{C},$$

since in the quotient, x satisfies $x^2 = -1$, which is precisely the defining relation of $i \in \mathbb{C}$.

Remark (Structural Position)*.* Quotient rings of polynomial rings are the coordinate rings of affine varieties, which are studied in depth in the next chapter. The passage

$$k[x_1, \ldots, x_n] \;\longrightarrow\; k[x_1, \ldots, x_n]/I$$

is the algebraic encoding of restricting from all of $\mathbb{A}^n$ to the variety $V(I)$.

Chapter 4

Linear Algebra

linear-algebra

Algebra $\rightarrow$ **Linear Algebra** $\rightarrow$ Formalizing Algebraic Structures

Remark (Toolkit — Vector Space Preliminaries). **Concept family:** The concrete foundations for $\mathbf{F}^n$ before the abstract vector space axioms are introduced.

Atomic items in this section:

- Definition: List
- Definition: $\mathbf{F}^n$
- Definition: Addition in $\mathbf{F}^n$
- Theorem: Addition Is Commutative in $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$
- Definition: Additive Inverse in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Key predicate:

$$\text{ListEquality}((x_1, \dots, x_n), (y_1, \dots, y_m)) \;:\equiv\; n = m \;\wedge\; \forall k \le n\ (x_k = y_k).$$

Definition (List)

Definition 4.0.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Remark (Standard quantified statement).

$$\text{List}(L) \iff \exists n \in \mathbb{N} \cup \{0\} : \big(\text{Length}(L) = n \wedge L = (x_1, x_2, \dots, x_n)\big)$$

Remark (Definition predicate reading).

$$\text{List}(L) \iff \exists n \in \mathbb{N}_0, \exists (x_1, \dots, x_n) : L = (x_1, \dots, x_n)$$

Remark (Interpretation). A list is the primary mechanism for discussing ordered finite sets of vectors or scalars. Unlike a set, the structural integrity of a list depends on both the identity and the position of its members.

- **Order Sensitivity**: The lists (a, b) and (b, a) are distinct unless $a = b$.
- **Finiteness**: By Axler's convention, a list must have a finite number of elements. Infinite ordered collections are categorized as sequences.
- **Empty List**: A list of length $n = 0$ is called the empty list, denoted by $()$.

Remark (Dependencies).

- Function
- Whole Numbers

Definition ($\mathbf{F}^n$)

Definition 4.0.2 ($\mathbf{F}^n$)**.** Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Remark (Standard quantified statement).

$$\begin{aligned} &x \in \mathbf{F}^n \\ &\iff \quad \exists x_1, \ldots, x_n \in \mathbf{F} \text{ such that } x = (x_1, \ldots, x_n). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Fn}(\mathbf{F}, n) \ :\equiv \ \mathbf{F}^n = \{(x_1, \ldots, x_n) : x_k \in \mathbf{F}\}.$$

Remark (Interpretation). An element of $\mathbf{F}^n$ is an ordered list of n scalars from $\mathbf{F}$. The order matters, and the list is completely specified by its coordinates.

Remark (Dependencies).

- Definition: List

Definition (Addition in $\mathbf{F}^n$)

Definition 4.0.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Standard quantified statement).

$$\begin{aligned} &\forall x, y \in \mathbf{F}^n, \\ &\quad x = (x_1, \ldots, x_n) \wedge y = (y_1, \ldots, y_n) \ \Rightarrow \ x + y = (x_1 + y_1, \ldots, x_n + y_n). \end{aligned}$$

Remark (Interpretation). Addition in $\mathbf{F}^n$ is defined coordinatewise: add the first coordinates, then the second coordinates, and so on through the nth coordinates.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Theorem (Addition Is Commutative in $\mathbf{F}^n$)

Theorem 4.0.4 (Addition Is Commutative in $\mathbf{F}^n$). *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbf{F}^n,\ x + y = y + x.$$

Remark (Interpretation). Because addition in $\mathbf{F}^n$ is coordinatewise and scalar addition in $\mathbf{F}$ is commutative, vector addition in $\mathbf{F}^n$ is commutative.

Remark (Dependencies).

- Definition: Addition in $\mathbf{F}^n$

Definition (Zero Vector in $\mathbf{F}^n$)

Definition 4.0.5 (Zero Vector in $\mathbf{F}^n$). The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Standard quantified statement).

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Interpretation). The zero vector is the vector whose every coordinate is the scalar 0 in $\mathbf{F}$. It is the additive identity for $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Definition (Additive Inverse in $\mathbf{F}^n$)

Definition 4.0.6 (Additive Inverse in $\mathbf{F}^n$). Let

$$x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \ldots, -x_n).$$

Remark (Standard quantified statement).

$$\forall x \in \mathbf{F}^n,$$
$$x = (x_1, \ldots, x_n) \ \Rightarrow\ -x = (-x_1, \ldots, -x_n).$$

Remark (Interpretation). The additive inverse in $\mathbf{F}^n$ is formed coordinatewise by taking the additive inverse of each coordinate in $\mathbf{F}$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$

Definition (Scalar Multiplication in $\mathbf{F}^n$)

Definition 4.0.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Remark (Standard quantified statement).

$$\forall \lambda \in \mathbf{F}\ \forall x \in \mathbf{F}^n,$$
$$x = (x_1, \ldots, x_n) \ \Rightarrow\ \lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Remark (Interpretation). Scalar multiplication in $\mathbf{F}^n$ is defined coordinatewise: multiply each coordinate of the vector by the scalar λ.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Remark (Toolkit — Vector Spaces). **Concept family:** The abstract vector space axioms and their immediate consequences.

Atomic items in this section:

- Definition: Vector Space (10 axioms)
- Definition: Vector and Point
- Definition: Real Vector Space
- Definition: Complex Vector Space
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

- Theorem: $a0 = 0$
- Theorem: $(-1)v = -v$

Key predicate:

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot).$$

The 10 axioms are closure, commutativity, associativity, additive identity, additive inverse, scalar closure, both distributive laws, scalar associativity, and the scalar identity.

Definition (Vector Space)

Definition 4.0.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Remark (Standard quantified statement).

$$\begin{aligned}
&V \text{ is a vector space over } \mathbf{F} \\
&\quad \iff \quad V \text{ is closed under addition and scalar multiplication,} \\
&\quad\quad \text{addition on } V \text{ is commutative and associative,} \\
&\quad\quad V \text{ has an additive identity and additive inverses,} \\
&\quad\quad \text{and scalar multiplication satisfies the two distributive laws,} \\
&\quad\quad \text{the associative law for scalar multiplication, and the identity law.}
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot)$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot) \iff \text{at least one of the ten axioms fails for some } u, v, w \in V,\ a, b \in \mathbf{F}.$$ ∎

Remark (Interpretation). A vector space is a set whose elements can be added together and multiplied by scalars, with these operations satisfying the natural algebraic rules familiar from $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$ (motivating example)
- Definition: Addition in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Definition (Vector and Point)

Definition 4.0.9 (Vector and Point)**.** An element of a vector space V is called a **vector**.
When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Remark (Standard quantified statement).

$$v \in V \implies v \text{ is a vector,}$$
$$x \in \mathbf{F}^n \implies x \text{ is a point of } \mathbf{F}^n.$$

Remark (Interpretation). The word *vector* emphasizes the algebraic role of an element of V. In $\mathbf{F}^n$, the same object can also be viewed geometrically as a point with coordinates in $\mathbf{F}$.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Real Vector Space)

Definition 4.0.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Remark (Standard quantified statement).

$$V \text{ is a real vector space} \iff V \text{ is a vector space over } \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealVectorSpace}(V) :\equiv \mathrm{VectorSpace}(V, \mathbb{R}, +, \cdot).$$

Remark (Interpretation). In a real vector space, the scalars used in scalar multiplication are real numbers.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Complex Vector Space)

Definition 4.0.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Remark (Standard quantified statement).

$$V \text{ is a complex vector space} \iff V \text{ is a vector space over } \mathbb{C}.$$

Remark (Definition predicate reading).

$$\mathrm{ComplexVectorSpace}(V) \;:\equiv\; \mathrm{VectorSpace}(V, \mathbb{C}, +, \cdot).$$

Remark (Interpretation). In a complex vector space, the scalars used in scalar multiplication are complex numbers.

Remark (Dependencies).

- Definition: Vector Space

Remark (Notation). For each positive integer m, the notation $\mathbf{F}^m$ denotes the vector space of all lists of length m of elements of $\mathbf{F}$.

The notation $\mathbf{F}^\infty$ denotes the set of all infinite lists

$$(x_1, x_2, x_3, \dots)$$

with each $x_j \in \mathbf{F}$, equipped with coordinatewise addition and coordinatewise scalar multiplication.

Remark (Coordinates in Vector Spaces). A pointwise proof is available when vectors are given explicitly by coordinates, as in $\mathbf{F}^n$, because the operations are defined coordinatewise. In an arbitrary vector space, vectors need not come with coordinates, so proofs must use the vector space axioms rather than coordinate calculations.

Theorem (Unique Additive Identity)

Theorem 4.0.12 (Unique Additive Identity)**.** *A vector space has a unique additive identity. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{If } 0, 0' \in V \text{ each satisfy}\\ &\qquad \forall v \in V,\ v + 0 = v \quad \text{and} \quad \forall v \in V,\ v + 0' = v,\\ &\text{then } 0 = 0'.\end{aligned}$$

Remark (Interpretation). Although the vector space axioms assert the existence of an additive identity, they force that identity to be unique.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Unique Additive Inverse)

Theorem 4.0.13 (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse. Go to proof.*

Remark (Standard quantified statement).

$$\forall v \in V,\ \text{if } w, u \in V \text{ satisfy } v + w = 0 \text{ and } v + u = 0,$$
$$\text{then } w = u.$$

Remark (Interpretation). The vector space axioms guarantee that each vector has an additive inverse, and that inverse is determined uniquely by the requirement that the sum be 0.

Remark (Notation). If $v \in V$, then the unique additive inverse of v is denoted by $-v$.

If $w, v \in V$, then

$$w - v \ := \ w + (-v).$$

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Identity

Theorem ($0v = 0$)

Theorem 4.0.14 ($0v = 0$)**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ 0v = 0.$$

Remark (Interpretation). Multiplying any vector by the scalar 0 gives the zero vector. The 0 on the left is the scalar zero in $\mathbf{F}$; the 0 on the right is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem ($a0 = 0$)

Theorem 4.0.15 ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbf{F},\ a0 = 0.$$

Remark (Interpretation). Every scalar sends the zero vector to the zero vector. The 0 on the right of a is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem ($(-1)v = -v$)

Theorem 4.0.16 ($(-1)v = -v$)**.** *For every* $v \in V$*,*

$$(-1)v = -v.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ (-1)v = -v.$$

Remark (Interpretation). Scalar multiplication by -1 produces the additive inverse of a vector. This justifies the notation $-v$ for the additive inverse: it is literally (-1) times v.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

Remark (Toolkit — Vector Subspaces). **Concept family:** Subspaces, their sums, and direct sums.

Atomic items in this section:

- Definition: Subspace
- Theorem: Conditions for a Subspace
- Definition: Sum of Subspaces
- Theorem: Sum of Subspaces Is the Smallest Containing Subspace
- Definition: Direct Sum
- Theorem: Condition for a Direct Sum
- Theorem: Direct Sum of Two Subspaces

Key predicates:

Subspace(U, V), ClosedUnderAddition(U), ClosedUnderScalarMultiplication$(U, \mathbf{F})$, Direct

Key test: U is a subspace $\iff$ $0 \in U$, U closed under $+$, U closed under scalar multiplication.

Definition (Subspace)

Definition 4.0.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\quad \iff \quad U \subseteq V \text{ and } U \text{ is a vector space over } \mathbf{F} \\ &\qquad \text{under the operations inherited from } V. \end{aligned}$$

Remark (Definition predicate reading)*.*

$$\text{Subspace}(U, V) \;:\equiv\; U \subseteq V \;\wedge\; \text{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Negated quantified statement)*.*

$$\neg\,\text{Subspace}(U, V) \iff U \not\subseteq V \;\vee\; \neg\,\text{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Interpretation)*.* A subspace is a subset of a vector space that is closed under the vector-space operations and therefore forms a vector space in its own right.

Remark (Dependencies)*.*

- Definition: Vector Space

Theorem (Conditions for a Subspace)

Theorem 4.0.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$*;*

(ii) for all $u, v \in U$, one has $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\quad \iff \quad 0 \in U \\ &\qquad \wedge\; \text{ClosedUnderAddition}(U) \\ &\qquad \wedge\; \text{ClosedUnderScalarMultiplication}(U, \mathbf{F}). \end{aligned}$$

Remark (Failure modes). U fails to be a subspace via exactly one of three routes: it omits the zero vector, it contains two vectors whose sum falls outside U, or it contains a vector and a scalar whose product falls outside U.

Remark (Contrapositive quantified statement).

$$\neg\ \text{Subspace}(U, V) \iff 0 \notin U \vee \neg\ \text{ClosedUnderAddition}(U) \vee \neg\ \text{ClosedUnderScalarMultiplication}(U,$$

Remark (Interpretation). To check that a subset is a subspace, it suffices to verify the three conditions. The remaining seven vector space axioms are inherited automatically from V since U uses the same operations. The three conditions are minimal: condition (i) ensures U is non-empty; (ii) and (iii) ensure the inherited operations stay within U.

Remark (Dependencies).

- Definition: Subspace
- Definition: Vector Space

Definition (Sum of Subspaces)

Definition 4.0.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} &v \in U_1 + \cdots + U_m \\ &\quad \iff \quad \exists u_1 \in U_1 \ \cdots\ \exists u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{SumOfSubspaces}(W, \{U_1, \ldots, U_m\}, V) \;:\equiv\; W = U_1 + \cdots + U_m.$$

Remark (Interpretation). The sum of subspaces consists of all vectors that can be built by adding together one vector from each subspace. Vectors from different subspaces interact; the representation $v = u_1 + \cdots + u_m$ need not be unique.

Remark (Dependencies).

- Definition: Subspace

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)

Theorem 4.0.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Subspace}(U_1 + \cdots + U_m,\, V),\\
&U_j \subseteq U_1 + \cdots + U_m \quad \text{for } j = 1, \ldots, m,\\
&\forall W \subseteq V\ \Big[\big(W \text{ is a subspace of } V \wedge U_1 \subseteq W \wedge \cdots \wedge U_m \subseteq W\big)\\
&\qquad\qquad \Rightarrow\ U_1 + \cdots + U_m \subseteq W\Big].
\end{aligned}$$

Remark (Interpretation). The sum is the smallest subspace containing all the given subspaces: it is their join in the lattice of subspaces of V. Any subspace containing all of $U_1, \ldots, U_m$ must contain every sum $u_1 + \cdots + u_m$, so it contains the sum subspace.

Remark (Dependencies).

- Definition: Sum of Subspaces
- Theorem: Conditions for a Subspace

Definition (Direct Sum)

Definition 4.0.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \ldots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \ldots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Remark (Standard quantified statement).

$$\begin{aligned}
&V = U_1 \oplus \cdots \oplus U_m\\
&\quad \iff \quad V = U_1 + \cdots + U_m\\
&\qquad \wedge\ \forall v \in V\ \exists!\, u_1 \in U_1\ \cdots\ \exists!\, u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m.
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \ :\equiv\ V = U_1 \oplus \cdots \oplus U_m.$$

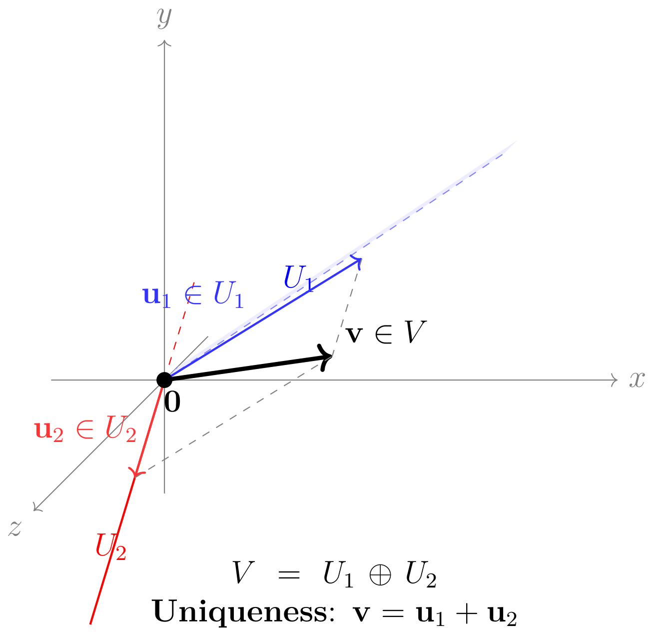
Remark (Negated quantified statement).

$$\neg\,\text{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \iff \exists v \in V\ \exists \text{ two distinct decompositions of } v.$$

Remark (Interpretation). A direct sum is a sum in which every vector has a unique decomposition into pieces coming from the given subspaces. The uniqueness condition is the structural rigidity that makes direct sums so useful — it means the subspaces contribute independently, with no redundancy.

Remark (Dependencies).

- Definition: Sum of Subspaces



$$V \;=\; U_1 \oplus U_2$$
Uniqueness: $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2$
Trivial Intersection:
$$U_1 \cap U_2 \;=\; \{\mathbf{0}\}$$

Theorem (Condition for a Direct Sum)

Theorem 4.0.22 (Condition for a Direct Sum)**.** *Let V be a vector space over* $\mathbf{F}$*, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \dots, u_m \in U_m$ and

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &V = U_1 \oplus \cdots \oplus U_m \\ &\quad\Longleftrightarrow \quad \forall u_1 \in U_1 \;\cdots\; \forall u_m \in U_m \\ &\qquad\quad \big(u_1 + \cdots + u_m = 0 \Rightarrow u_1 = \cdots = u_m = 0\big). \end{aligned}$$

Remark (Interpretation)*.* A sum is direct exactly when the zero vector has only the trivial decomposition as a sum of vectors from the given subspaces. This is the practical test: rather than verifying uniqueness for every $v \in V$, it suffices to check uniqueness at $v = 0$.

Remark (Dependencies).

- Definition: Direct Sum
- Theorem: Unique Additive Identity

Theorem (Direct Sum of Two Subspaces)

Theorem 4.0.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U + W = U \oplus W \\ &\iff \quad U \cap W = \{0\}. \end{aligned}$$

Remark (Contrapositive quantified statement).

$$U \cap W \neq \{0\} \iff U + W \neq U \oplus W.$$

If the two subspaces share a nonzero vector, the sum is not direct.

Remark (Interpretation). For two subspaces, directness is equivalent to saying they overlap only in the zero vector. This gives a simple geometric test: draw both subspaces and check whether their intersection is just the origin. The result does not generalise to three or more subspaces — it is possible for $U_1 \cap U_2$, $U_1 \cap U_3$, and $U_2 \cap U_3$ to all equal $\{0\}$ yet $U_1 + U_2 + U_3$ still fail to be a direct sum.

Remark (Dependencies).

- Theorem: Condition for a Direct Sum
- Definition: Direct Sum

Chapter 5

Lattice Order

lattice-order

Linear Algebra $\to$ **Lattice Order** $\to$ Set Algebra

Why Lattice Theory Gets Its Own Home

Remark (Why lattice theory gets its own home). Lattice theory begins with a simple observation: in many ordered structures, two elements have a natural least upper bound and a natural greatest lower bound. Once those operations exist everywhere, order starts behaving like algebra.

Remark (Where the first pass already lives). The first introduction to lattices is already in the lattice section of the order notes. That is the right place for the first encounter, because it grows directly out of partial orders, suprema, and infima.

Remark (Why return later). Once lattices stop being examples and start becoming the main character, the story gets richer. A fuller lattice-theory chapter will naturally gather topics such as:

- distributive, modular, and Boolean lattices;
- complete lattices and closure operators;
- sublattices, ideals, and filters;
- Galois connections and adjoint situations;
- fixed-point theorems such as Knaster–Tarski;
- the lattice viewpoint behind topologies, sigma-algebras, and domain theory.

Remark (Why this matters for the rest of the curriculum). This chapter is a breadcrumb because lattice order quietly reappears all over the place. Power sets form Boolean lattices, sigma-algebras behave like closure-stable lattice structures, open sets form a join-rich order, and fixed-point methods show up in logic, measure, computation, and analysis. Giving lattice order its own future home makes those later echoes easier to recognize.

5.1 Ordered Sets

5.2 Ordered Sets

Remark (Toolkit: Ordered Sets)*.* Order is a relation, not a property. A set carries no intrinsic ordering; an order is an additional structure imposed on the set by specifying which pairs of elements stand in the ordering relation. The same underlying set can support many distinct orders simultaneously. The atomic notions developed here are:

- the **partial order** — the minimal framework in which order makes sense;
- the **strict partial order** — the asymmetric companion to the partial order, and the native language of open-ball conditions and limits;
- the **total order** — the specialisation in which every pair of elements is comparable.

Definition (Partial Order)

Definition 5.2.1 (Partial Order)**.** A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \implies a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \implies a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Remark (Standard quantified statement)*.*

$$\text{PartialOrder}(S, \leq) \;:=\; \big(\forall a \in S,\; a \leq a\big) \wedge \big(\forall a, b \in S,\; a \leq b \wedge b \leq a \Rightarrow a = b\big) \wedge \big(\forall a, b, c \in S,\; a \leq b \wedge b \leq$$

Remark (Negated quantified statement)*.*

$$\neg\,\text{PartialOrder}(S, \leq) \;:=\; \big(\exists a \in S,\; a \not\leq a\big) \vee \big(\exists a, b \in S,\; a \leq b \wedge b \leq a \wedge a \neq b\big) \vee \big(\exists a, b, c \in S,\; a \leq b \wedge b \leq$$

Remark (Interpretation)*.* Order is not an intrinsic quality of a set's elements — it is an external binary relation. The three axioms together enforce that the relation be a consistent, cycle-free, directed comparison. Reflexivity says every element is at least as large as itself. Anti-symmetry is the formal engine behind squeeze arguments: if $a \leq b$ and $b \leq a$ then a and b cannot be distinct. Transitivity ensures that the ordering chains: comparisons can be composed without breaking consistency. The same set of objects can carry many distinct partial orders simultaneously — standard, reverse, or lexicographic — each defining a different relational geometry.

Remark (Dependencies)*.*

- Partial Order

Definition

Definition 5.2.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \Longrightarrow \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \Longrightarrow a < c$.

Remark (Standard quantified statement)*.*

$$\mathrm{StrictPartialOrder}(S,<) \;:=\; \big(\forall a \in S,\ \neg(a < a)\big) \wedge \big(\forall a,b \in S,\ a < b \Rightarrow \neg(b < a)\big) \wedge \big(\forall a,b,c \in S,\ a <$$

Remark (Interpretation)*.* The strict order is the companion of the non-strict order, eliminating reflexivity and replacing anti-symmetry with the stronger asymmetry. Every strict partial order determines a partial order by $a \leq b \iff (a < b \vee a = b)$, and every partial order determines a strict partial order by $a < b \iff (a \leq b \wedge a \neq b)$. In metric spaces and analysis, strict orders are the natural language of open conditions: open balls, open intervals, and limit definitions all use $<$ rather than $\leq$ to exclude boundary contact.

Remark (Dependencies)*.*

- Partial Order

Definition (Total Order)

Definition 5.2.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

Remark (Standard quantified statement)*.*

$$\mathrm{TotalOrder}(S,\leq) \;:=\; \mathrm{PartialOrder}(S,\leq) \;\wedge\; \forall a,b \in S,\ (a < b) \oplus (a = b) \oplus (b < a),$$

where $\oplus$ denotes exclusive or. Equivalently, comparability suffices: $\forall a, b \in S,\ a \leq b \vee b \leq a$.

Remark (Interpretation)*.* A partial order permits elements to be incomparable: it is possible that neither $a \leq b$ nor $b \leq a$. A total order removes this possibility entirely — every pair of elements is forced to be comparable. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ under their standard orderings are totally ordered fields. By contrast, the power set $\mathcal{P}(X)$ of any set X with more than one element, ordered by inclusion, is typically not a total order: the singletons $\{a\}$ and $\{b\}$ with $a \neq b$ are incomparable.

Remark (Dependencies)*.*

Partial order, strict partial order.

5.3 Chains and Antichains

Remark (Toolkit: Chains and Antichains). A poset can contain subsets whose elements are either completely aligned or entirely incomparable. These two extremes are formalised as chains and antichains. They measure the “linearity” and “width” of the order structure respectively.

Definition (Chain)

Definition 5.3.1 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \lor \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Remark (Standard quantified statement).

$$\mathrm{Chain}(C, S, \leq) \ \coloneqq \ C \subseteq S \ \land \ \forall x, y \in C, \ x \leq y \lor y \leq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Chain}(C, S, \leq) \ \coloneqq \ \exists x, y \in C, \ x \nleq y \land y \nleq x.$$

Remark (Interpretation). A chain is a subset in which the partial order behaves as a total order: no two elements are left incomparable. The entire real line $\mathbb{R}$ under its standard ordering is itself a chain. Any finite totally ordered set — a ranking, a sequence of nested intervals, a linearly ordered list of stages — is a chain. Chains model the “ladder” structure of order: there is a clear direction from smaller to larger, and the path from any element to any other is unambiguous.

Remark (Dependencies).

Partial order, total order.

Definition (Antichain)

Definition 5.3.2 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \land \neg(y \leq x).$$

Remark (Standard quantified statement).

$$\mathrm{Antichain}(A, S, \leq) \ \coloneqq \ A \subseteq S \ \land \ \forall x, y \in A, \ x \neq y \Rightarrow x \nleq y \land y \nleq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Antichain}(A, S, \leq) \ \coloneqq \ \exists x, y \in A, \ x \neq y \land (x \leq y \lor y \leq x).$$

Remark (Interpretation). An antichain is a subset in which the partial order makes no comparisons at all: every pair of distinct elements is incomparable. In the power set $\mathcal{P}(X)$ ordered by inclusion, the collection of all subsets of a fixed cardinality k forms an antichain — no subset of size k contains another. Antichains measure the “width” of a poset: Dilworth's theorem states that the minimum number of chains needed to cover a finite poset equals the maximum size of an antichain. In higher-dimensional structures such as product orders, large antichains arise naturally and are the primary tool for understanding the lateral spread of the order.

Remark (Dependencies).

Partial order.

5.4 Partial and Total Maps

Remark (Toolkit: Partial and Total Maps). Before studying maps that preserve order structure, the foundational function types are recorded: a partial map need not be defined everywhere, while a total map is defined on every input. These are the base layer upon which the order-theoretic notions rest.

Definition (Partial Map)

Definition 5.4.1 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\ \forall y_1, y_2 \in B, \quad (x, y_1) \in f \ \wedge\ (x, y_2) \in f \implies y_1 = y_2.$$

Remark (Standard quantified statement).

$$\mathrm{PartialMap}(f, A, B) \ :=\ f \subseteq A \times B \wedge \forall x \in A,\ \forall y_1, y_2 \in B,\ (x, y_1) \in f \wedge (x, y_2) \in f \Rightarrow y_1 = y_2.$$

Remark (Interpretation). A partial map assigns to each element of A at most one element of B. Its domain of definition $\mathrm{dom}(f) = \{x \in A : \exists y \in B,\ (x, y) \in f\}$ may be a proper subset of A. Partial maps arise in order theory when a function is well-defined only on elements satisfying some order-theoretic condition — for example, on elements that admit an upper bound.

Definition

Definition 5.4.2 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\ \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

Remark (Standard quantified statement).

$$\mathrm{TotalMap}(f, A, B) \ :=\ \mathrm{PartialMap}(f, A, B) \ \wedge\ \forall x \in A,\ \exists y \in B,\ (x, y) \in f.$$

Remark (Dependencies).

Partial map, Function.

5.5 Maps Between Ordered Sets

Remark (Toolkit: Maps Between Ordered Sets). The structure-preserving maps between posets form a hierarchy of increasing strength. An order-preserving map carries the direction of the order forward. An order-embedding carries it in both directions. An order-isomorphism is a surjective order-embedding — a perfect structural identification between two posets.

Definition (Order-Preserving Map)

Definition 5.5.1 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Remark (Standard quantified statement).

$$\text{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \Rightarrow \varphi(x) \leq_Q \varphi(y).$$

Remark (Negated quantified statement).

$$\neg\,\text{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \exists x, y \in P,\; x \leq_P y \wedge \varphi(x) \not\leq_Q \varphi(y).$$

Remark (Interpretation). An order-preserving map carries the direction of the order from P into Q: whenever x is at or below y in P, $\varphi(x)$ is at or below $\varphi(y)$ in Q. The implication is one-directional — order-preserving maps need not reflect the order back. Two order-preserving maps compose to an order-preserving map, making order-preserving maps the morphisms of the category of posets.

Remark (Dependencies).

Partial order.

Definition (Order-Embedding)

Definition 5.5.2 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Remark (Standard quantified statement).

$$\text{OrderEmbedding}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

Remark (Interpretation). An order-embedding both preserves and reflects the order: the order structure of P is faithfully copied into Q without distortion. Unlike mere preservation, reflection means that order-relations between images in Q genuinely originate from order-relations in P. Every order-embedding is injective (Lemma 5.5.4) and identifies P order-isomorphically with its image in Q.

Remark (Dependencies).

Order-preserving map.

Definition (Order-Isomorphism)

Definition 5.5.3 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \;\coloneqq\; \mathrm{OrderEmbedding}(\varphi, P, Q) \;\wedge\; \forall q \in Q,\; \exists p \in P,\; \varphi(p) = q.$$

Remark (Interpretation). An order-isomorphism is a bijection that perfectly translates the order of P into Q and back. Two order-isomorphic posets are order-theoretically identical: any statement about P expressed purely in terms of $\leq_P$ translates to an equally valid statement about Q under φ. Order-isomorphism is the correct notion of equality for posets.

Remark (Dependencies).
Order-embedding.

Lemma

Lemma 5.5.4 (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \implies \forall x, y \in P,\; \varphi(x) = \varphi(y) \Rightarrow x = y.$$

Remark (Interpretation). Injectivity follows automatically from the biconditional in the embedding definition. If $\varphi(x) = \varphi(y)$, the image is simultaneously $\leq$ and $\geq$ itself in Q; order-reflection pulls this back to P, and anti-symmetry then forces $x = y$.

Remark (Dependencies).
Order-embedding, anti-symmetry of partial orders.

5.6 The Covering Relation

Remark (Toolkit: The Covering Relation). The covering relation identifies the irreducible steps of a partial order: y covers x when $x < y$ and there is nothing strictly between them. In a finite poset the entire order is determined by its covering relation, which is encoded exactly by the Hasse diagram.

Definition (Covering Relation)

Definition 5.6.1 (Covering Relation). Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \neg\big(\exists z \in P,\; x < z < y\big).$$

Remark (Standard quantified statement).

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \forall z \in P,\; x < z \leq y \Longrightarrow z = y.$$

Remark (Negated quantified statement).

$$\neg(x \prec y) \;\coloneqq\; x \not< y \;\vee\; \exists z \in P,\; x < z < y.$$

Remark (Interpretation). The covering relation captures the direct steps of the order — the comparisons that cannot be decomposed further. In a finite poset every strict comparison $x < y$ can be resolved into a finite chain of covers $x = x_0 \prec x_1 \prec \cdots \prec x_n = y$, so the covering relation alone determines the full order. The Hasse diagram of a finite poset is exactly the directed graph of the covering relation. In a dense order such as $\mathbb{Q}$ or $\mathbb{R}$, no element covers any other: between any two elements there is always a third.

Remark (Dependencies).

Partial order, strict partial order.

Lemma

Lemma 5.6.2 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \iff \big(\forall x, y \in P,\ x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)\big) \iff \big(\forall x, y \in P,\ x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi($$

Remark (Interpretation). In a finite poset the order relation, the strict order, and the covering relation each determine the others completely. This lemma shows that an order-isomorphism can be detected at any one of these three levels. The covering characterisation is the most useful in practice: since Hasse diagrams encode exactly the covering relation, two finite posets are order-isomorphic precisely when their diagrams are isomorphic as directed graphs.

Remark (Dependencies).

Order-isomorphism, covering relation.

Proposition

Proposition 5.6.3 (Hasse Diagram Characterization for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements. Go to proof.*

Remark (Standard quantified statement).

$$P \cong Q \iff \exists \varphi : P \xrightarrow{\sim} Q,\ \forall x, y \in P,\ x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y).$$

Remark (Interpretation). The Hasse diagram is a complete invariant of the order structure of a finite poset. Checking order-isomorphism between finite posets reduces to a graph isomorphism problem on the covering graphs — two finite posets with non-isomorphic Hasse diagrams cannot be order-isomorphic.

Remark (Dependencies).

Characterizations of order-isomorphisms for finite posets, covering relation.

5.7 Duality

Remark (Toolkit: Duality). Every partially ordered set has a dual obtained by reversing all order relations. Any statement that holds in every poset therefore holds in every dual poset, which means its order-reversed version also holds universally. This is the Duality Principle — the formal engine behind phrases like "by duality, the corresponding result for infima follows."

Definition (Dual Ordered Set)

Definition 5.7.1 (Dual Ordered Set)**.** Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Remark (Standard quantified statement).

$$x \leq_{P^{\mathrm{op}}} y \coloneqq y \leq_P x.$$

Remark (Interpretation). Dualising a poset exchanges the roles of upper and lower: every upper bound becomes a lower bound, every supremum becomes an infimum, and every maximal element becomes a minimal element. In a finite poset the Hasse diagram of P^{op} is obtained from that of P by turning it upside down. The dual of a total order is again a total order.

Remark (Dependencies).

Partial order.

Proposition

Proposition 5.7.2 (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set. Go to proof.*

Remark (Standard quantified statement). Every universally valid statement about partially ordered sets has a universally valid order-dual statement.

Remark (Interpretation). The Duality Principle eliminates the need to re-prove results whose dual is already established. Any theorem about upper bounds, suprema, maximal elements, or top elements automatically yields a theorem about lower bounds, infima, minimal elements, or bottom elements by applying the principle to P^{op}.

Remark (Dependencies).

Dual ordered set.

5.8 Extremal Elements

Remark (Toolkit: Extremal Elements). A poset may contain elements that sit at the boundary of the order. There are two distinct senses of "extreme": an element can be extreme *locally* (nothing in the poset exceeds it, though it may be incomparable with some elements) or extreme *globally* (it dominates or is dominated by every element). The definitions below make these distinctions precise, proceeding from global extremes of the whole poset, through local extremes of a subset, to the proposition that finite posets always contain the local kind.

Definition (Bottom Element)

Definition 5.8.1 (Bottom Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\perp \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \perp \leq x.$$

If a bottom element exists it is unique; it is denoted $\perp_P$ or simply $\perp$.

Remark (Standard quantified statement).

$$\mathrm{Bottom}(\perp, P, \leq) \;:=\; \perp \in P \;\wedge\; \forall x \in P,\; \perp \leq x.$$

Remark (Interpretation). A bottom element is below every element of the poset without exception. In $(\mathcal{P}(X), \subseteq)$ the bottom element is $\varnothing$. The natural numbers have a bottom element; the integers do not. An antichain with more than one element has no bottom element, since no member is below any other.

Remark (Dependencies).

Partial order.

Definition (Top Element)

Definition 5.8.2 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Remark (Standard quantified statement).

$$\mathrm{Top}(\top, P, \leq) \;:=\; \top \in P \;\wedge\; \forall x \in P,\; x \leq \top.$$

Remark (Interpretation). Dual to the bottom element via the Duality Principle. In $(\mathcal{P}(X), \subseteq)$ the top element is X itself.

Remark (Dependencies).

Bottom element, duality principle.

Definition

Definition 5.8.3 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Remark (Standard quantified statement)*.*

$$\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; a \leq x \Rightarrow a = x.$$

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; \exists x \in Q,\; a < x,$$

that is, some element of Q strictly exceeds a.

Remark (Interpretation)*.* A maximal element asserts only that nothing in Q strictly dominates it. It need not be comparable with every element of Q, so a poset can have multiple maximal elements simultaneously. In domain theory, maximal elements correspond to fully-specified or total objects.

Remark (Dependencies)*.*

Partial order.

Definition

Definition 5.8.4 (Minimal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Remark (Standard quantified statement)*.*

$$\mathrm{Minimal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; x \leq a \Rightarrow x = a.$$

Remark (Interpretation)*.* Dual to maximal element via the Duality Principle.

Remark (Dependencies)*.*

Maximal element, duality principle.

Definition

Definition 5.8.5 (Greatest Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Remark (Standard quantified statement)*.*

$$\mathrm{Greatest}(m, Q, \leq) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; x \leq m.$$

Remark (Interpretation). A greatest element is a maximal element that is additionally comparable with every element of Q. When $\max Q$ exists, $\mathrm{Max}(Q) = \{\max Q\}$, but the converse fails: a unique maximal element need not dominate every element.

Remark (Dependencies).

Maximal element.

Definition

Definition 5.8.6 (Least Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Remark (Standard quantified statement).

$$\mathrm{Least}(m, Q, \leq) \;:=\; m \in Q \;\wedge\; \forall x \in Q,\; m \leq x.$$

Remark (Interpretation). Dual to greatest element.

Remark (Dependencies).

Greatest element, duality principle.

Proposition

Proposition 5.8.7 (Every Nonempty Finite Poset Has a Maximal Element). *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.* *Go to proof.*

Remark (Standard quantified statement).

$$P \text{ finite},\; Q \subseteq P,\; Q \neq \varnothing \implies \mathrm{Max}(Q) \neq \varnothing.$$

$$\forall x \in P,\; \exists y \in \mathrm{Max}(P),\; x \leq y.$$

Remark (Interpretation). Finiteness is essential. The collection of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element. In a finite poset one can always ascend from any element to a ceiling. This property is the finite shadow of Zorn's Lemma and underlies ascending chain arguments throughout algebra and order theory.

Remark (Dependencies).

Maximal element, partial order.

5.9 Lifting and Co-Lifting

Remark (Toolkit: Lifting and Co-Lifting). A poset that lacks a bottom or top element can be given one by adjoining a new universal bound. Lifting adjoins a bottom; co-lifting adjoins a top. These constructions are dual to each other and are used throughout domain theory, where a bottom element represents an undefined state.

Definition (Lifting)

Definition 5.9.1 (Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Remark (Standard quantified statement).

$$P_\perp = P \cup \{\perp\}, \quad \perp \notin P, \quad \forall x \in P_\perp,\ \perp \leq x, \quad \forall x, y \in P,\ x \leq_{P_\perp} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Lifting adjoins a universal lower bound to a poset that may not have one. In domain theory $\perp$ represents an undefined, divergent, or unspecified state. Any set S viewed as an antichain can be lifted to the *flat poset* $S_\perp$, in which all elements of S are incomparable above $\perp$.

Remark (Dependencies).
Partial order, Union, bottom element.

Definition

Definition 5.9.2 (Co-Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

Remark (Standard quantified statement).

$$P^\top = P \cup \{\top\}, \quad \top \notin P, \quad \forall x \in P^\top,\ x \leq \top, \quad \forall x, y \in P,\ x \leq_{P^\top} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Dual to lifting via the Duality Principle: co-lifting adjoins a universal upper bound. The co-lifting of an antichain produces a flat poset in which all elements are incomparable below $\top$.

Remark (Dependencies).
Lifting, top element, duality principle.

5.10 Consistency

Remark (Toolkit: Consistency). Two elements of a poset are consistent when they share a common upper bound — when they can be jointly extended to some element above both of them. In lattice-theoretic settings this is automatic; in general posets it is a genuine condition.

Definition

Definition 5.10.1 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge\ y \leq z.$$

Remark (Standard quantified statement).

$$\mathrm{Consistent}(x, y, S, \leq) \;:=\; \exists z \in S,\; x \leq z \wedge y \leq z.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Consistent}(x, y, S, \leq) \;:=\; \forall z \in S,\; x \not\leq z \vee y \not\leq z.$$

Remark (Interpretation). Consistency asserts that x and y are compatible — there is some element of S above both. In a lattice where binary joins always exist, consistency is automatic: take $z = x \vee y$. In a general poset the condition is nontrivial and may fail. In domain theory, consistency captures the idea that two partial computations describe information that can be combined without contradiction.

Remark (Dependencies).
Partial order.

5.11 Down-Sets and Up-Sets

Remark (Toolkit: Down-Sets and Up-Sets). A down-set is a subset of a poset that is closed downward: if an element belongs to it, everything below that element belongs to it too. An up-set is the dual notion — closed upward. Down-sets are the order ideals of lattice theory; up-sets are the order filters. The collection of all down-sets of a poset, ordered by inclusion, is itself an important poset.

Definition (Down-Set)

Definition 5.11.1 (Down-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\; \forall y \in P, \quad y \leq x \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{DownSet}(Q, P, \leq) \;:=\; Q \subseteq P \;\wedge\; \forall x \in Q,\; \forall y \in P,\; y \leq x \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DownSet}(Q, P, \leq) \;:=\; \exists x \in Q,\; \exists y \in P,\; y \leq x \wedge y \notin Q.$$

Remark (Interpretation). A down-set is a downward-closed region of the poset: once an element is included, the entire lower section below it is forced in. Down-sets generalise downward-closed intervals. The collection of all down-sets of P, denoted $\mathcal{O}(P)$ and ordered by inclusion, is itself a poset and in the finite case a distributive lattice. Birkhoff's representation theorem states that every finite distributive lattice arises this way.

Remark (Dependencies).
Partial order.

Definition (Up-Set)

Definition 5.11.2 (Up-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\; \forall y \in P, \quad x \leq y \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{UpSet}(Q, P, \leq) \;\coloneqq\; Q \subseteq P \;\wedge\; \forall x \in Q,\; \forall y \in P,\; x \leq y \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{UpSet}(Q, P, \leq) \;\coloneqq\; \exists x \in Q,\; \exists y \in P,\; x \leq y \wedge y \notin Q.$$

Remark (Interpretation). Dual to down-set via the Duality Principle: the up-sets of P are exactly the down-sets of P^{op}. In topology, filters are special up-sets closed under finite intersection and satisfying a directedness condition. The complement of a down-set is an up-set, and vice versa.

Remark (Dependencies).

Down-set, duality principle.

5.12 Families of Sets

Remark (Toolkit: Families of Sets). The power set of any set is naturally ordered by inclusion. Every collection of subsets — whether arising from algebra, topology, or logic — inherits this order. The key observation is that subsets correspond exactly to predicates via their characteristic functions, and that this correspondence is an order-isomorphism.

Remark (Families as ordered sets). Let X be a set. The *power set* $\mathcal{P}(X)$, consisting of all subsets of X, is ordered by inclusion: for $A, B \in \mathcal{P}(X)$,

$$A \leq B \iff A \subseteq B.$$

Any subcollection of $\mathcal{P}(X)$ inherits this order. Such a collection is called a *family of sets*.

Families of sets arise naturally when X carries additional structure. If X is a group G, the set $\mathrm{Sub}(G)$ of all subgroups and the set $\mathrm{NSub}(G)$ of all normal subgroups are each ordered by inclusion. If X is a vector space V, the set $\mathrm{Sub}(V)$ of all subspaces is ordered by inclusion. If X is a ring R, the sets of all subrings and of all ideals of R are similarly ordered. If $(X, \mathcal{T})$ is a topological space, the collections of open sets, closed sets, and clopen sets are each families of sets ordered by inclusion.

A particularly important example is the family $\mathcal{O}(P)$ of all down-sets of a partially ordered set P, ordered by inclusion.

Remark (Predicate formulation). The ordered set $(\mathcal{P}(X), \subseteq)$ admits an equivalent formulation in terms of predicates. A *predicate* on X is a function

$$p : X \to \{T, F\},$$

where T and F denote truth values. Let $\mathcal{P}^*(X)$ denote the set of all predicates on X, ordered by implication:

$$p \leq q \iff \{x \in X : p(x) = T\} \subseteq \{x \in X : q(x) = T\}.$$

Define the map

$$\Phi : \mathcal{P}^*(X) \to \mathcal{P}(X), \qquad \Phi(p) \coloneqq \{x \in X : p(x) = T\}.$$

Proposition

Proposition 5.12.1 (Predicates and Power Sets Are Order-Isomorphic)**.** *The map* $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ *is an order-isomorphism. Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\Phi, \mathcal{P}^*(X), \mathcal{P}(X)).$$

Remark (Interpretation). Every subset of X can be represented as the truth set of a predicate, and every predicate determines a subset. The correspondence is bijective and order-preserving in both directions: $p \leq q$ if and only if the truth set of p is contained in the truth set of q, which is exactly the inclusion order on $\mathcal{P}(X)$. This identification is fundamental in logic, where sets and their characteristic predicates are treated as interchangeable, and in computer science, where sets are often implemented via their membership tests.

Remark (Dependencies).

Order-isomorphism, partial order.

Chapter 6

Set Algebra

set-algebra

Lattice Order → **Set Algebra** → Formalizing Algebraic Structures

6.1 Set Algebras

6.2 Systems of Sets

Comparison of Set Systems

Property	Semiring	Ring	Algebra	σ-algebra	Topology
Contains $\varnothing$	✓	✓	✓	✓	✓
Contains X	—	—	✓	✓	✓
Closed under $\cap$ (finite)	✓	✓	✓	✓	✓
Closed under $\cup$ (finite)	—	✓	✓	✓	✓
Closed under $\cup$ (countable)	—	—	—	✓	—
Closed under $\cup$ (arbitrary)	—	—	—	—	✓
Closed under complement	—	—	✓	✓	—
Difference $A \setminus B$	finite disj. union	✓	✓	✓	—

The definitions and theorems of the previous sections have been about individual sets and their operations. A second level of structure emerges when one asks: which *collections* of sets are closed under given operations? This question is not abstract generality for its own sake. The three subjects that depend most heavily on this chapter — topology, measure theory, and functional analysis — each begin by specifying a particular type of collection on a given set X. A topology specifies which subsets are "open." A σ-algebra specifies which subsets are "measurable." In both cases, the definition is precisely a list of closure conditions: which set operations applied to members of the collection must produce another member.

Mastering these four structures here, as purely set-theoretic objects, means that when topology and measure theory are encountered, all the foundational work is already done. The task will be to understand the *geometric* or *analytic* meaning of the axioms, not to struggle with the set-theoretic mechanics.

Definition (Semiring of Sets)

Definition 6.2.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \ldots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Set Difference
- Union
- Pairwise Disjoint Family
- Finite and Infinite Sets

Example (Half-open intervals form a semiring). Let $X = \mathbb{R}$ and let $\mathcal{S}$ be the collection of all half-open intervals $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$, together with $\varnothing$. Then $\mathcal{S}$ is a semiring:

- $[a, b) \cap [c, d) = [\max(a, c), \min(b, d))$, which is again a half-open interval (or $\varnothing$);
- $[a, b) \setminus [c, d)$ is either $\varnothing$, $[a, c)$, $[d, b)$, or $[a, c) \sqcup [d, b)$ — in every case, a finite disjoint union of half-open intervals.

This semiring is the natural starting point for constructing Lebesgue measure: one first defines length on $\mathcal{S}$ (as $b - a$ for $[a, b)$), then extends to the generated ring and σ-algebra.

Remark (Why semirings appear in measure theory). Lebesgue measure is most naturally defined first on a semiring, because semirings have enough structure to state the countable additivity condition for a *premeasure*, yet are simple enough that explicit formulas for sets like $[a, b)$ are easy to write down. The general extension theorem (Carathéodory's theorem) then extends the premeasure to the full σ-algebra. This is why Kolmogorov & Fomin introduce semirings before algebras: they are the base case of the extension hierarchy.

Definition (Ring of Sets)

Definition 6.2.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Remark (Dependencies).

- Subset
- Empty Set
- Union
- Set Difference

Remark (Derived closure properties of a ring). A ring is closed under finitely many of each operation. Specifically, if $\mathcal{R}$ is a ring then:

- $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$ (intersection follows from set difference);
- $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{R}$ (symmetric difference follows from union and difference).

A ring is closed under all finite Boolean combinations of its elements.

Remark (Why "ring"). The name comes from algebra. Under the operations of symmetric difference $\triangle$ (playing the role of addition) and intersection $\cap$ (multiplication), a ring of sets is a commutative ring in the algebraic sense: $\triangle$ is commutative and associative with identity $\varnothing$, every element is its own inverse ($A \triangle A = \varnothing$), and $\cap$ distributes over $\triangle$.

Definition (Algebra of Sets)

Definition 6.2.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Remark (Dependencies).

- Subset
- Complement
- Union

Remark (Derived properties of an algebra)*.* An algebra is a ring that also contains X. Since $\varnothing = X^c$ and $X \in \mathcal{A}$, every algebra contains $\varnothing$. Closure under finite intersection follows: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan. An algebra is therefore closed under all finite Boolean operations: finite unions, finite intersections, complements, and differences.

Definition (σ-Algebra)

Definition 6.2.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space.*

Remark (Dependencies)*.*

- Subset
- Empty Set
- Complement
- Union
- Countable Set

Remark (Derived closure properties of a σ-algebra)*.* Every σ-algebra is an algebra (take the union of a finite family by padding with $\varnothing$). But a σ-algebra is additionally closed under countable intersections: by De Morgan,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{M}.$$

The σ in "σ-algebra" is notation for "countable" (from the German *Summe*), signalling the step up from finite to countable closure.

Remark (Why countable and not arbitrary)*.* A σ-algebra is closed under countable unions but not arbitrary ones. This is not a deficiency — it is the right condition for measure theory. Measure is defined to be countably additive: if $\{A_n\}$ is a countable disjoint family, then $\mu\big(\bigcup A_n\big) = \sum \mu(A_n)$. Extending to uncountable unions would force every set to have either measure zero or infinite measure (a useless theory). The countability restriction is what makes the theory nontrivial.

By contrast, a topology (defined below) is closed under *arbitrary* unions but only *finite* intersections. This is a different trade-off: open sets are defined by local conditions, and any union of open sets is open because being open is a pointwise property. Countable intersections of open sets are *not* required to be open; those are a separate class called G_δ sets.

Definition (Topology)

Definition 6.2.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Remark (Dependencies).

- Subset
- Empty Set
- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Finite and Infinite Sets

Remark (Why arbitrary unions but only finite intersections). A topology formalizes the idea of "nearness" without a distance function. The condition that arbitrary unions of open sets are open reflects the following: if a point x is near some set, and that set is a union of smaller sets, then x is near one of those smaller sets. This is a purely local condition and works for any union.

Finite intersections must be open because: if x lies in the interior of U_1 and also in the interior of U_2, it lies in the interior of $U_1 \cap U_2$. This argument works for finitely many sets (take the smallest neighborhood). For a countably infinite intersection, the "smallest neighborhood" might shrink to nothing — consider $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ in $\mathbb{R}$, which is not open. So infinite intersections of open sets are not required to be open.

Theorem (Intersection of σ-algebras is a σ-algebra)

Theorem 6.2.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. Go to proof.

Remark (Standard quantified statement). $\big(\forall \alpha \in I,\ \mathcal{M}_\alpha$ is a σ-algebra on $X\big) \to \bigcap_{\alpha \in I} \mathcal{M}_\alpha$ is a σ-algeb
Explicitly: $X \in \bigcap \mathcal{M}_\alpha$; $A \in \bigcap \mathcal{M}_\alpha \to A^c \in \bigcap \mathcal{M}_\alpha$; $\forall n,\ A_n \in \bigcap \mathcal{M}_\alpha \to \bigcup_n A_n \in \bigcap \mathcal{M}_\alpha$.

Remark (Dependencies).

- σ-Algebra

- Indexed Family of Sets
- Intersection

Remark (Proof). Check each axiom: (i) $X \in \mathcal{M}_\alpha$ for all α, so $X \in \bigcap_\alpha \mathcal{M}_\alpha$. (ii) If $A \in \mathcal{M}$ then $A \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}$. (iii) If $\{A_n\} \subseteq \mathcal{M}$ then $\{A_n\} \subseteq \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}$. All three conditions are verified. The same argument works for algebras, rings, and topologies (with appropriate closure conditions).

Definition (Generated σ-Algebra)

Definition 6.2.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) \;:=\; \bigcap \{\, \mathcal{M} \;\big|\; \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \,\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Remark (Dependencies).

- σ-Algebra
- Subset
- Intersection
- Intersection of σ-Algebras

Remark (Why the definition is valid). The set of σ-algebras on X containing $\mathcal{F}$ is nonempty: $\mathcal{P}(X)$ is a σ-algebra containing every subset of X. Theorem 6.2.6 guarantees their intersection is again a σ-algebra. So $\sigma(\mathcal{F})$ is always well-defined. The same construction works for generated algebras, generated rings, and (with a small modification for topologies) generated topologies.

Remark (How to think about generated structures). $\sigma(\mathcal{F})$ contains $\mathcal{F}$, and then contains everything that must be present in any σ-algebra that contains $\mathcal{F}$: complements of elements of $\mathcal{F}$, countable unions of elements of $\mathcal{F}$, countable unions of complements of elements of $\mathcal{F}$, and so on — iterating indefinitely. The generated σ-algebra is the "closure" of $\mathcal{F}$ under all σ-algebraic operations. In practice, one specifies $\mathcal{F}$ as a convenient collection (e.g., intervals) and then works with the full σ-algebra by knowing which operations are allowed.

Definition (Borel σ-Algebra)

Definition 6.2.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \;:=\; \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets*.

Remark (Dependencies).

- Topology
- Generated σ-Algebra

Example (F_σ and G_δ sets)*.* In a topological space X:

- A G_δ *set* is a countable intersection of open sets: $A = \bigcap_{n=1}^\infty U_n$ with each U_n open.
- An F_σ *set* is a countable union of closed sets: $A = \bigcup_{n=1}^\infty F_n$ with each F_n closed.

These names come from German: G for *Gebiet* (open set), δ for *Durchschnitt* (intersection); F for *Ferme* (closed), σ for *Summe* (union). Both G_δ and F_σ sets are Borel sets. They are dual: A is G_δ if and only if A^c is F_σ.

These classes appear in analysis and measure theory as the natural "next level" above open and closed sets. In $\mathbb{R}$: the set of continuity points of a monotone function is a G_δ set; the rational numbers $\mathbb{Q}$ are an F_σ set; the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are a G_δ set.

Remark (Transition)*.* The four structures — semiring, ring, algebra, σ-algebra — and the topology form a hierarchy of increasing closure strength. Measure theory builds from the bottom (semiring, ring) upward; topology is its own branch with a different closure pattern. The Borel σ-algebra bridges the two: it is generated by the topology, so it captures all sets describable by topological operations, and it is the standard "default" σ-algebra in any analytic context. Whenever a later chapter says "let f be a measurable function" without specifying the σ-algebra, it almost certainly means measurability with respect to the Borel σ-algebra.

6.3 The Power Set and Characteristic Functions

Bibliography